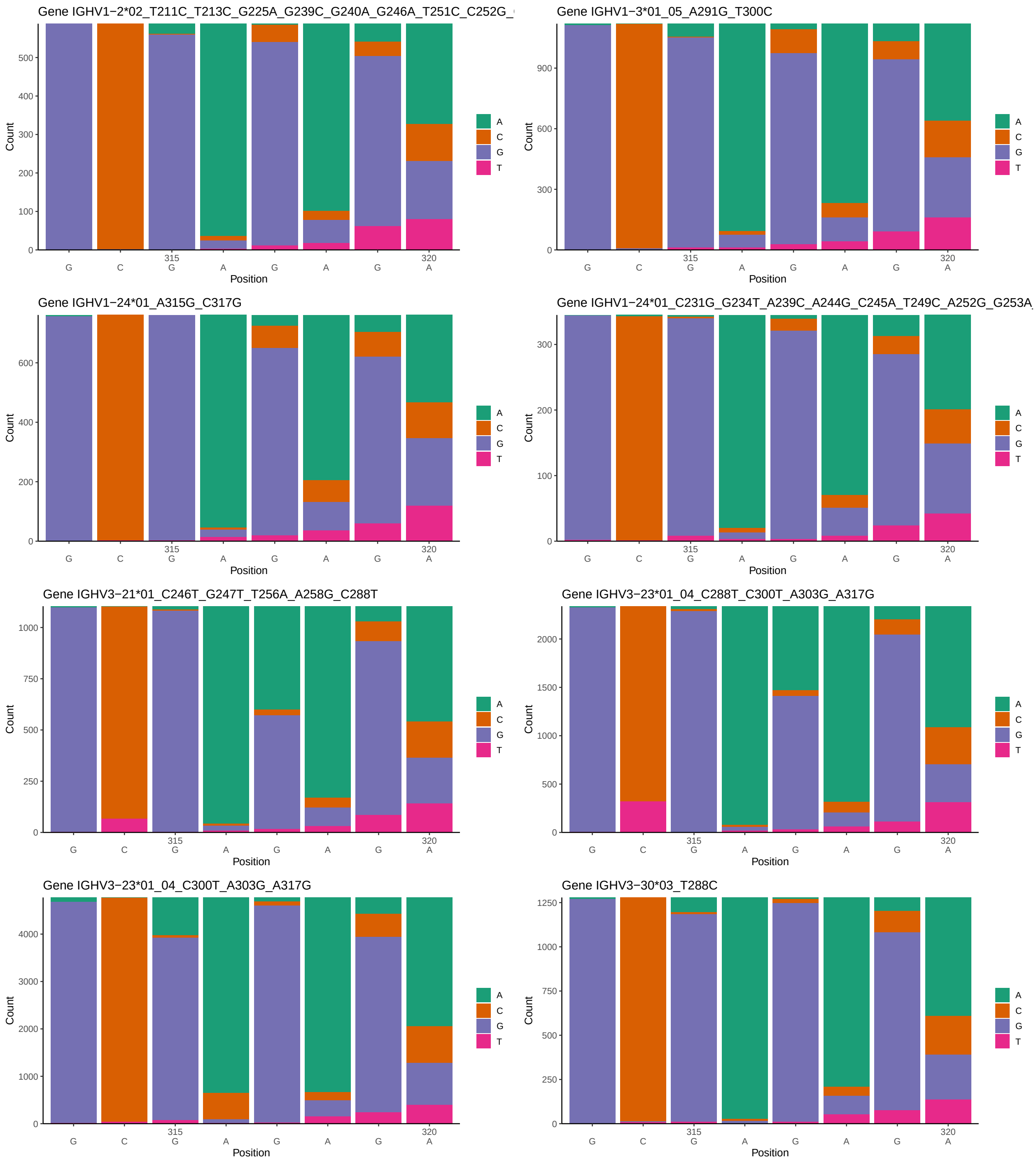
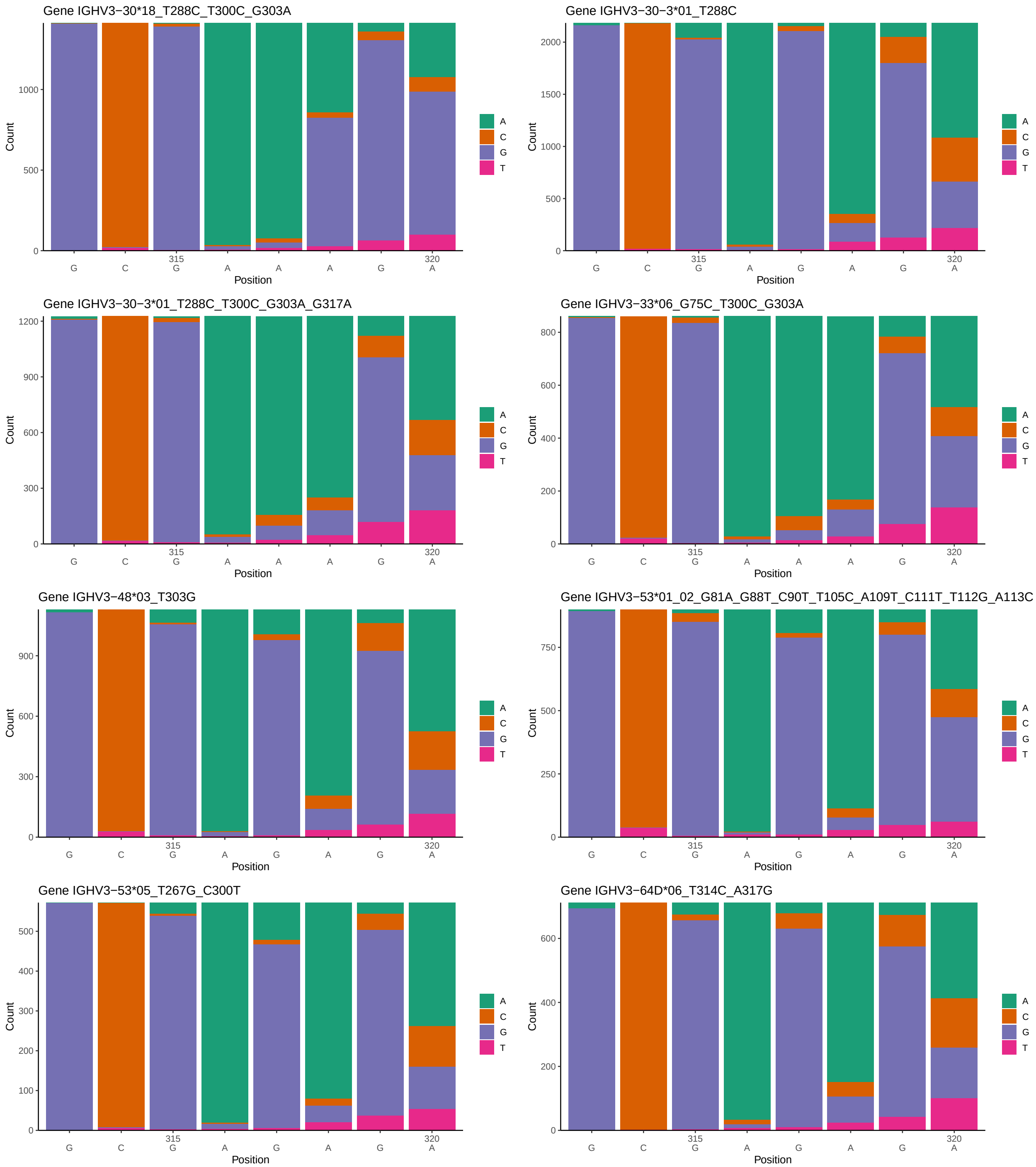
OGRDBstats Report

Contents

1 Novel sequence analysis

1.1 End-nucleotide composition

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Gene IGHV3−30−3*01_T288C_T300C_G303A_G317A
Gene IGHV3−33*06_G75C_T300C_G303A
Gene IGHV3−48*03_T303G
Gene IGHV3−53*01_02_G81A_G88T_C90T_T105C_A109T_C111T_T112G_A113C
Gene IGHV3−53*05_T267G_C300T
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G
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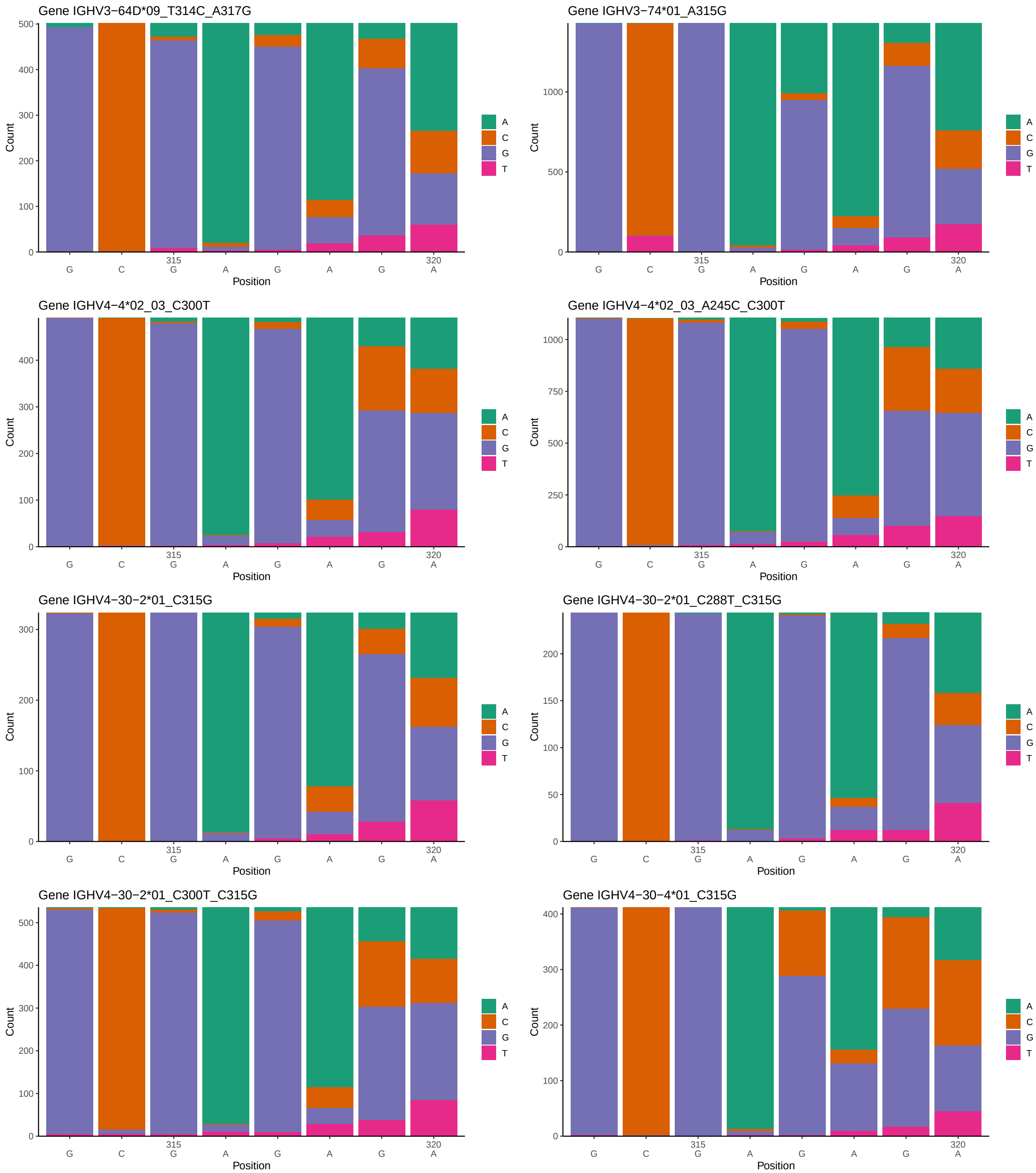
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Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−2*01_C288T_C315G
Gene IGHV4−30−2*01_C300T_C315G
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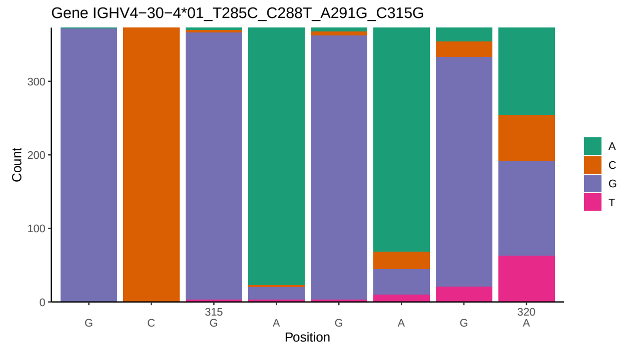
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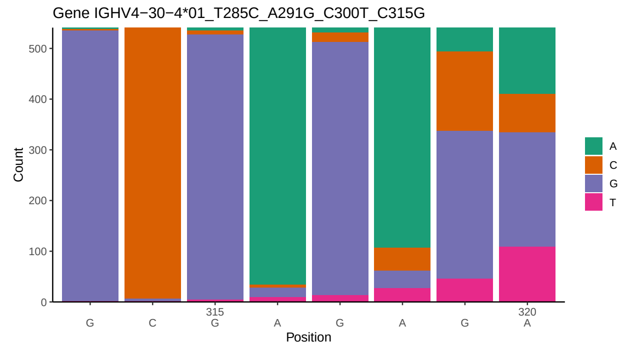
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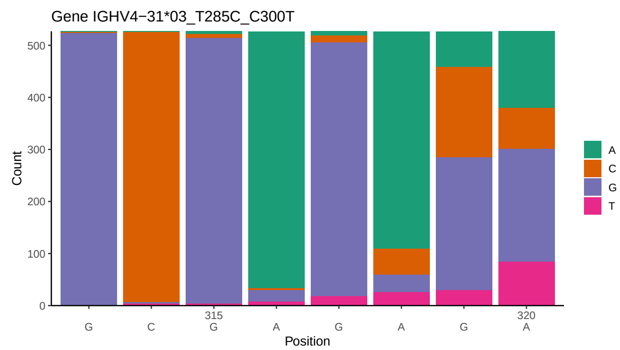
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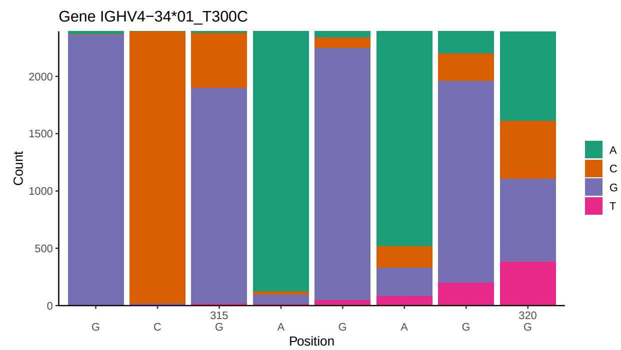
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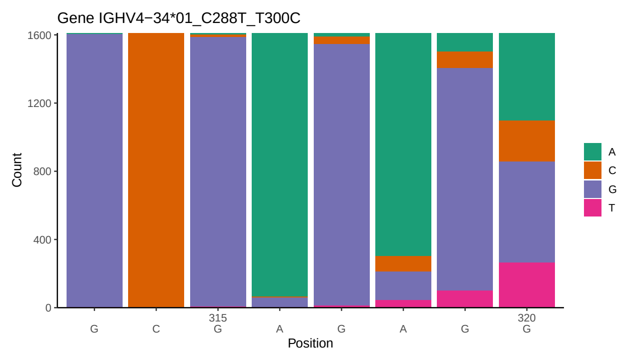
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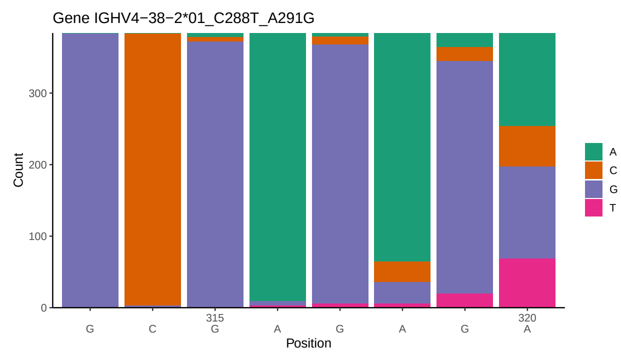
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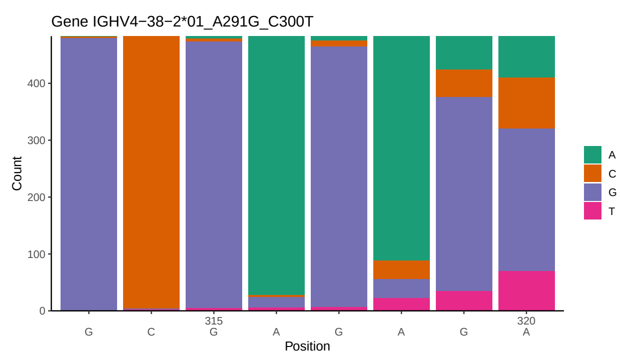
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C
G
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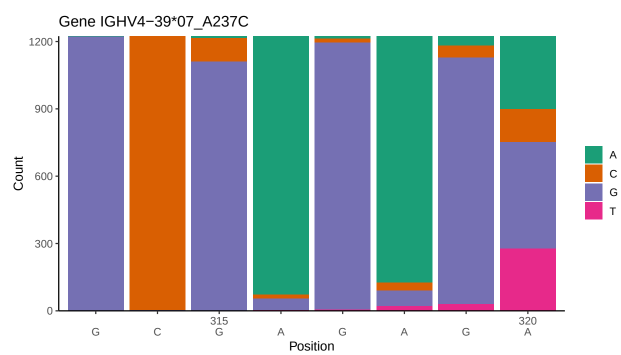
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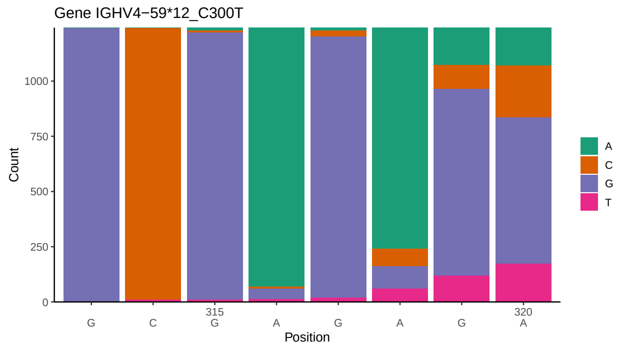

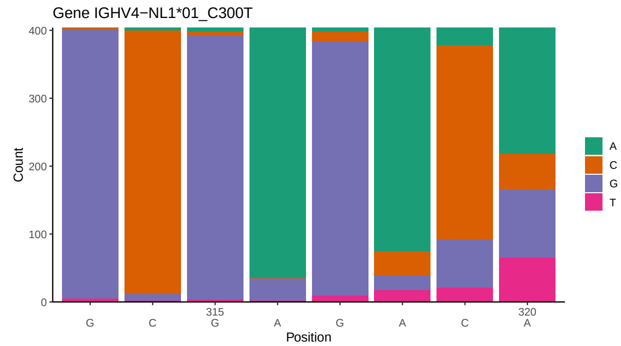

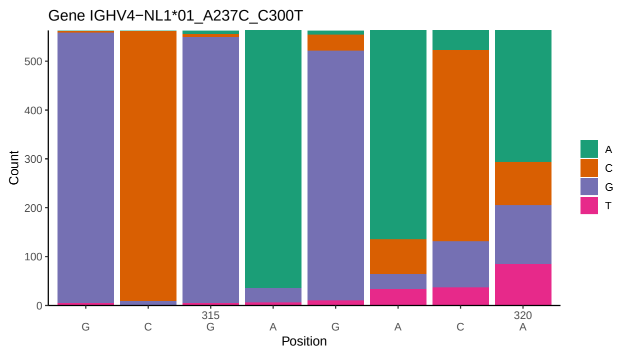

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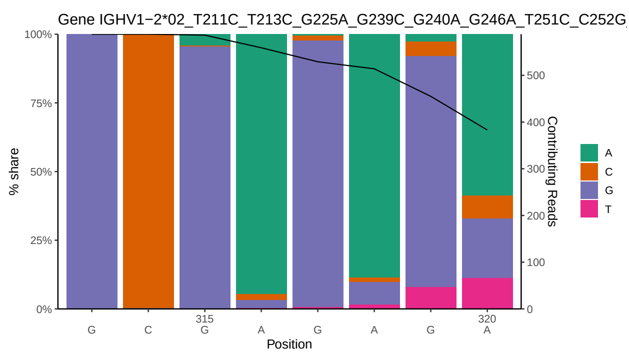

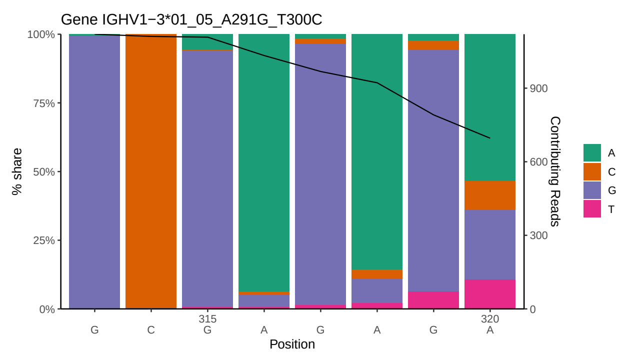

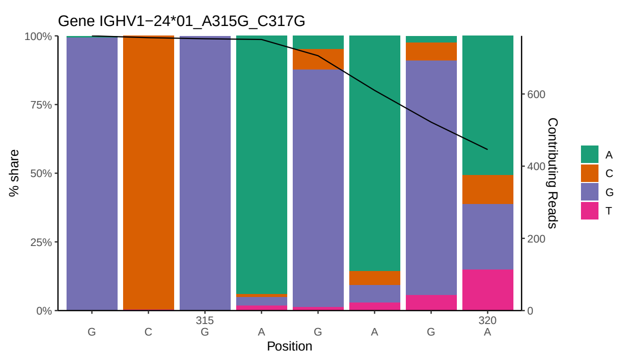

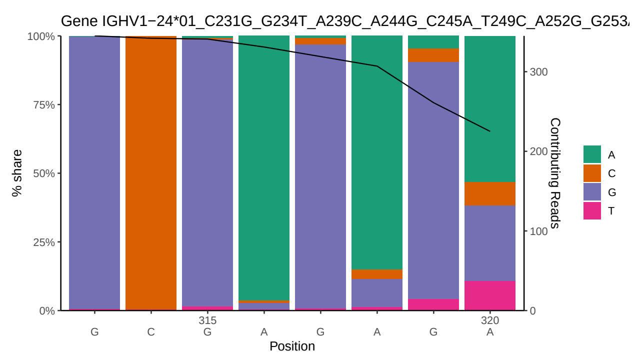

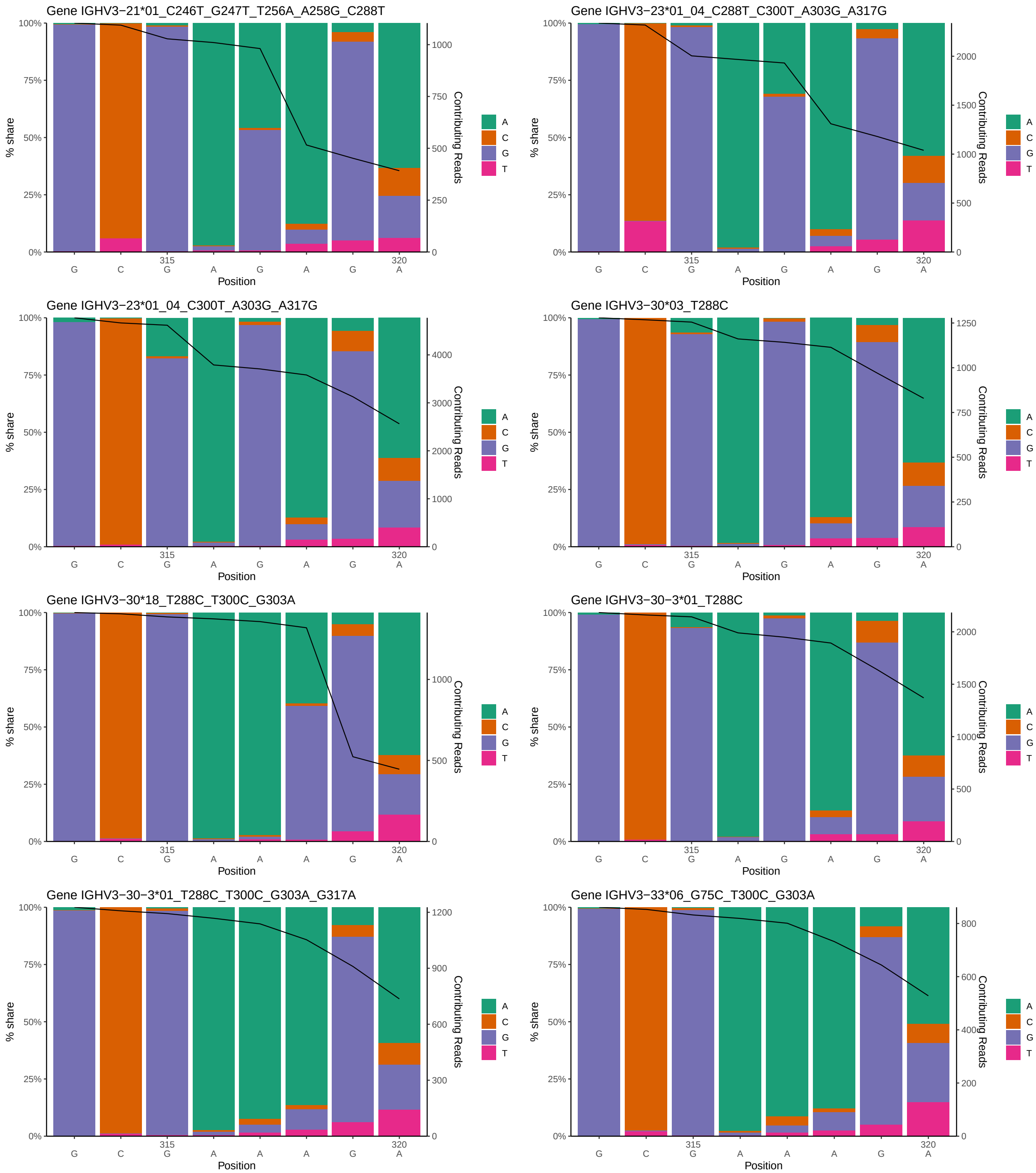
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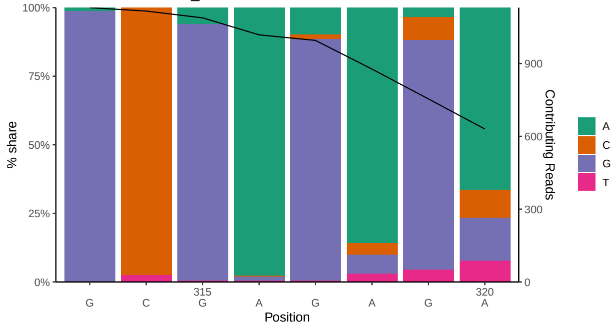
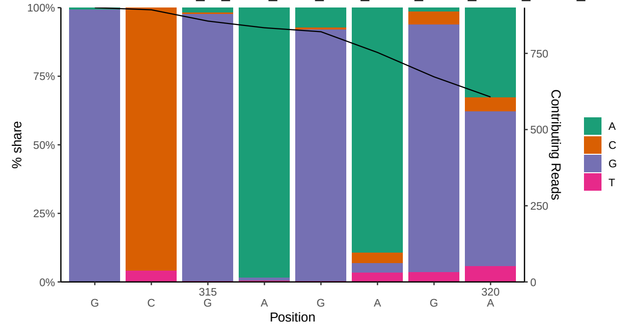
Gene IGHV3−23*01_04_C300T_A303G_A317G
Gene IGHV3−30*03_T288C
Gene IGHV3−30*18_T288C_T300C_G303A
Gene IGHV3−30−3*01_T288C
Gene IGHV3−30−3*01_T288C_T300C_G303A_G317A
Gene IGHV3−33*06_G75C_T300C_G303A
% share
0%
25%
50%
75%
100%
Contributing Reads
Position
315
320
G C G A G A G A
A
C
G
T

Gene IGHV3−48*03_T303G



Gene IGHV3−53*01_02_G81A_G88T_C90T_T105C_A109T_C111T_T112G_A113C



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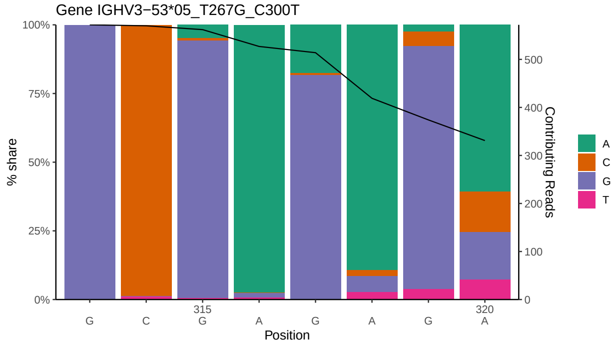

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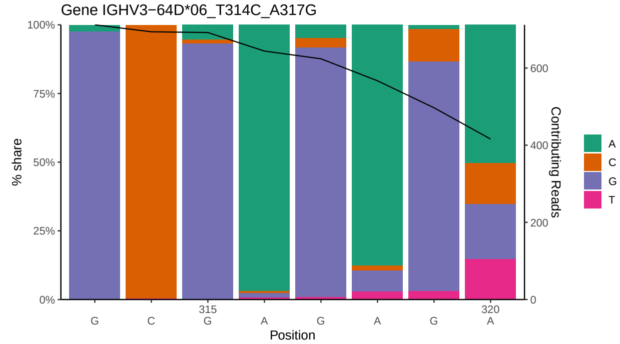

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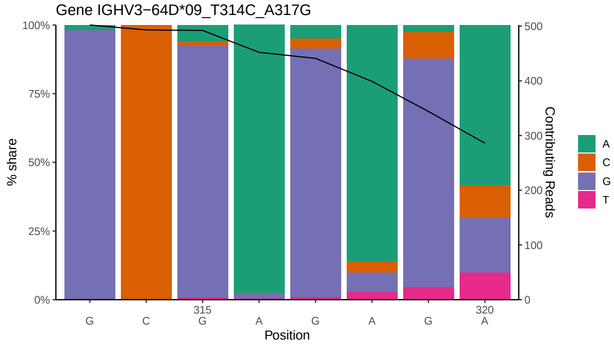

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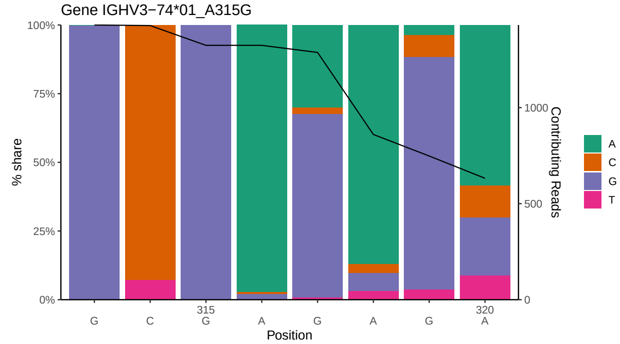
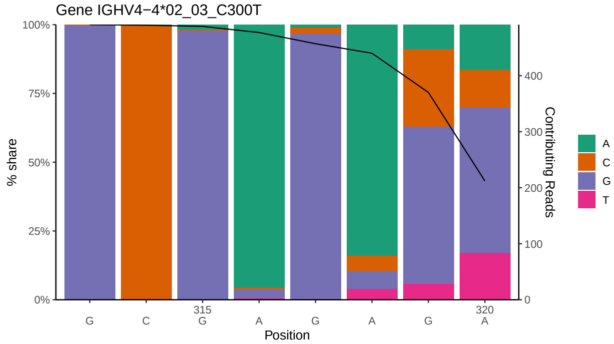


Gene IGHV4−4*02_03_C300T



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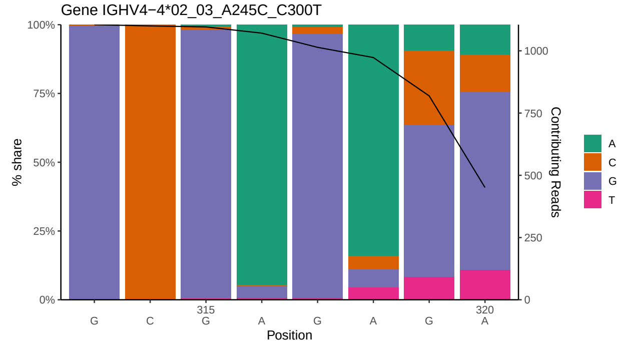

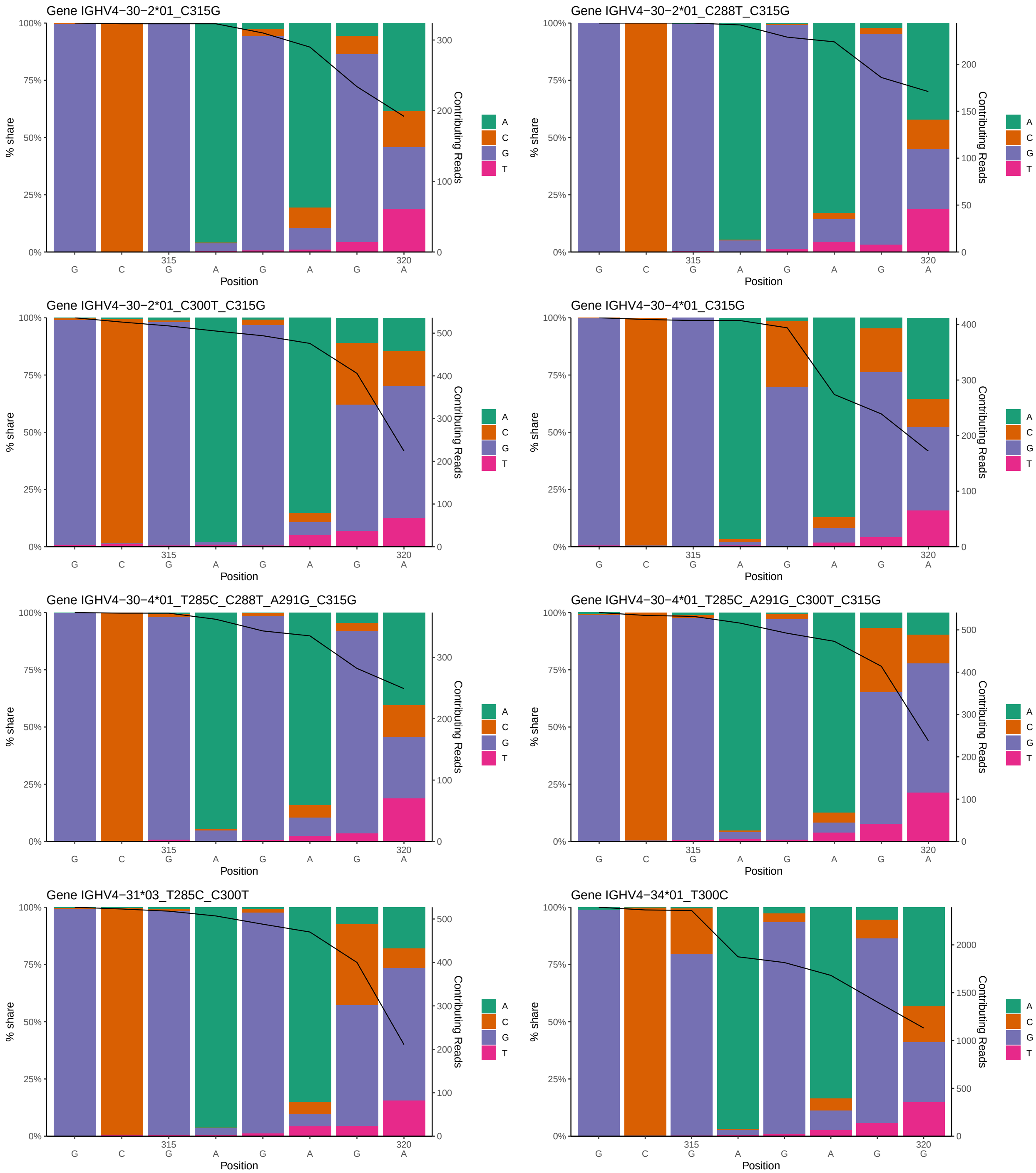
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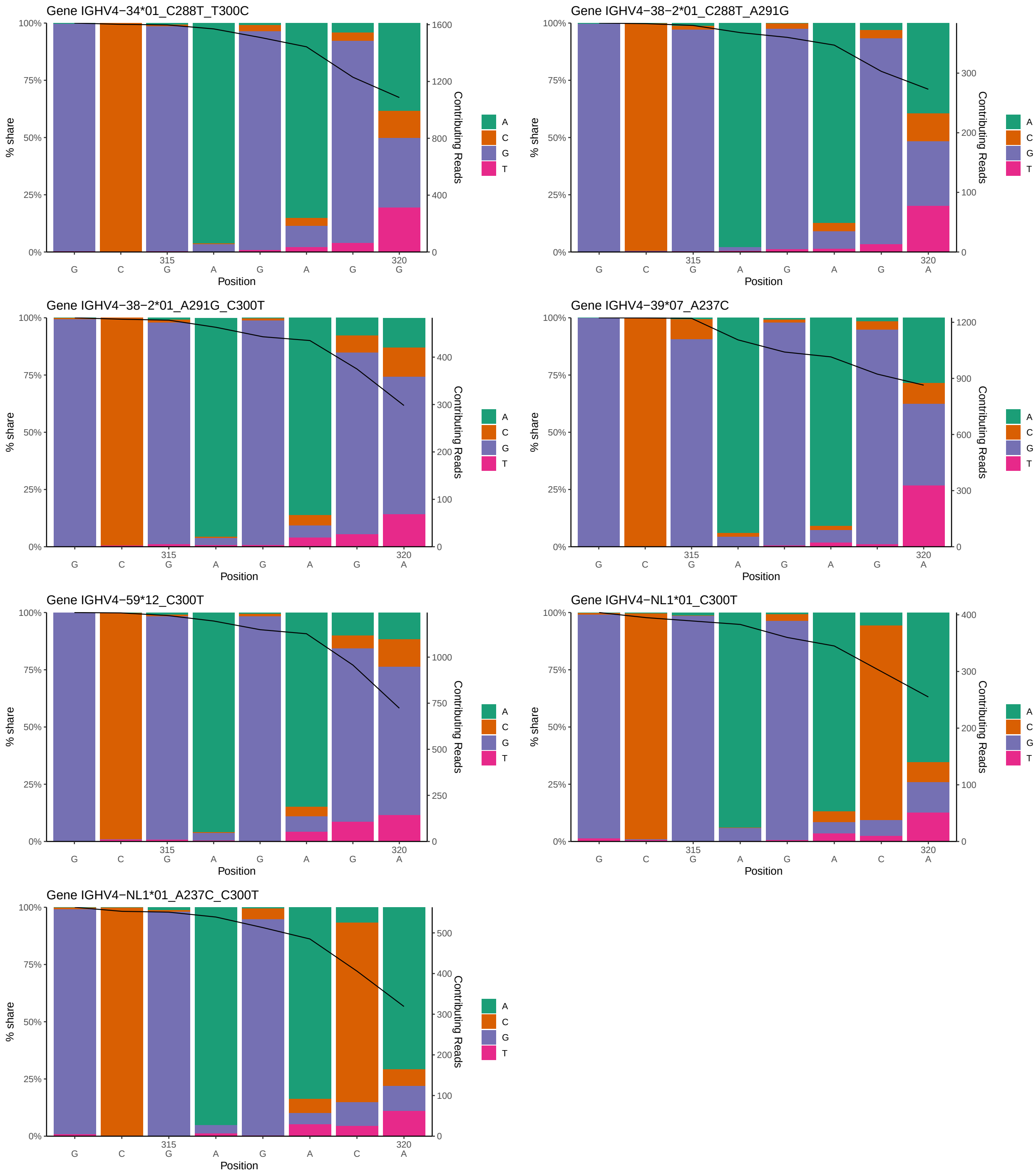
Gene IGHV4−30−2*01_C288T_C315G
Gene IGHV4−30−2*01_C300T_C315G
Gene IGHV4−30−4*01_C315G
Gene IGHV4−30−4*01_T285C_C288T_A291G_C315G
Gene IGHV4−30−4*01_T285C_A291G_C300T_C315G
Gene IGHV4−31*03_T285C_C300T
Gene IGHV4−34*01_T300C
% share
Contributing Reads
Position
A
C
G
T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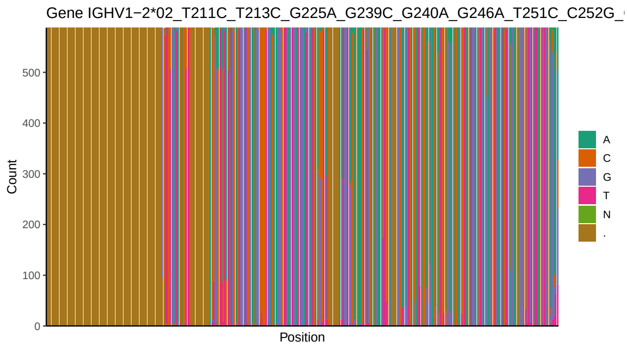
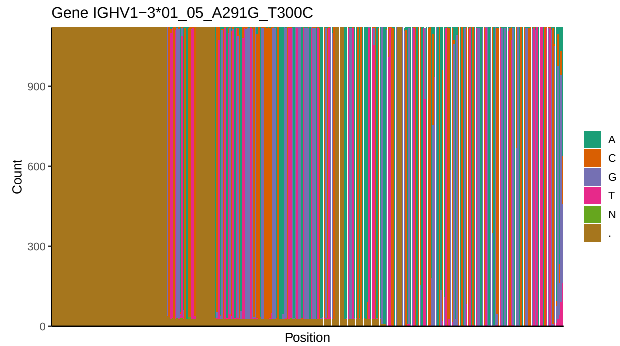
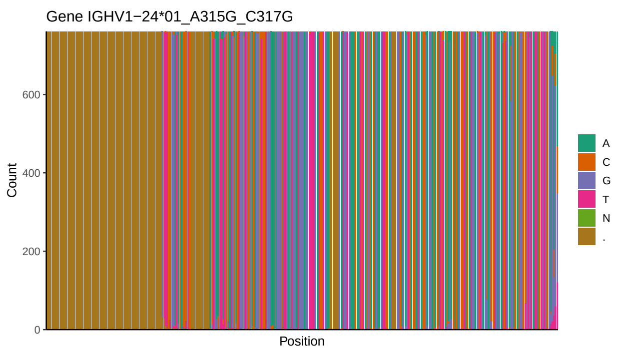
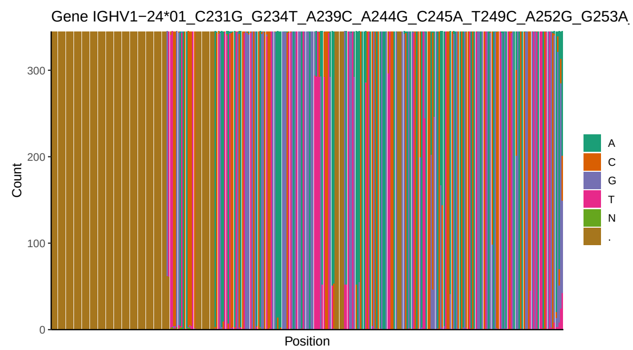
Gene IGHV4−38−2*01_C288T_A291G
Gene IGHV4−38−2*01_A291G_C300T
Gene IGHV4−39*07_A237C
Gene IGHV4−59*12_C300T
Gene IGHV4−NL1*01_C300T
Gene IGHV4−NL1*01_A237C_C300T
% share
Position
Contributing Reads
A
C
G
T

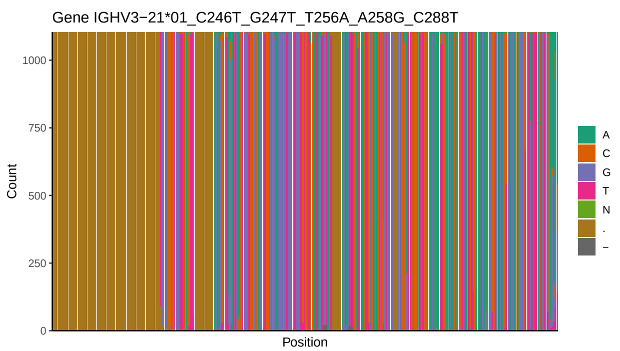
1.3 Whole-sequence composition of each assigned read

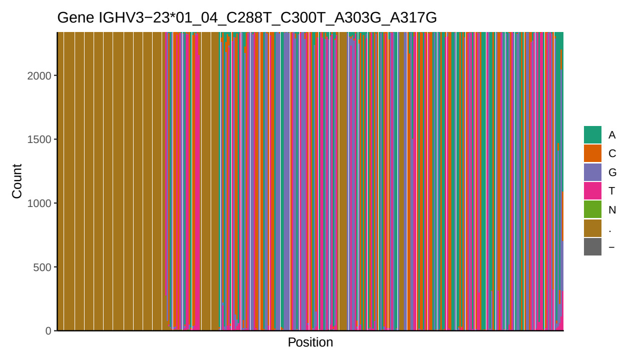


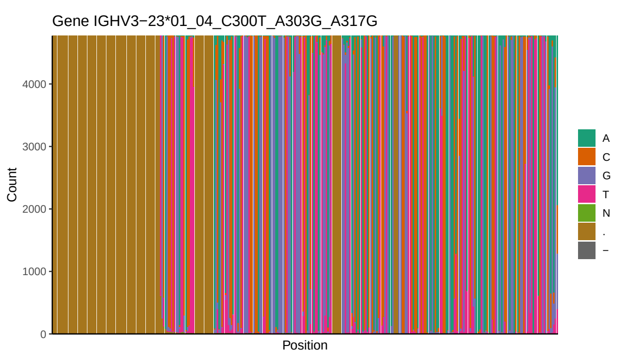


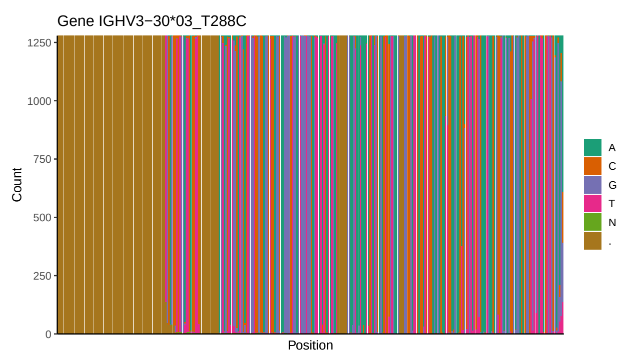


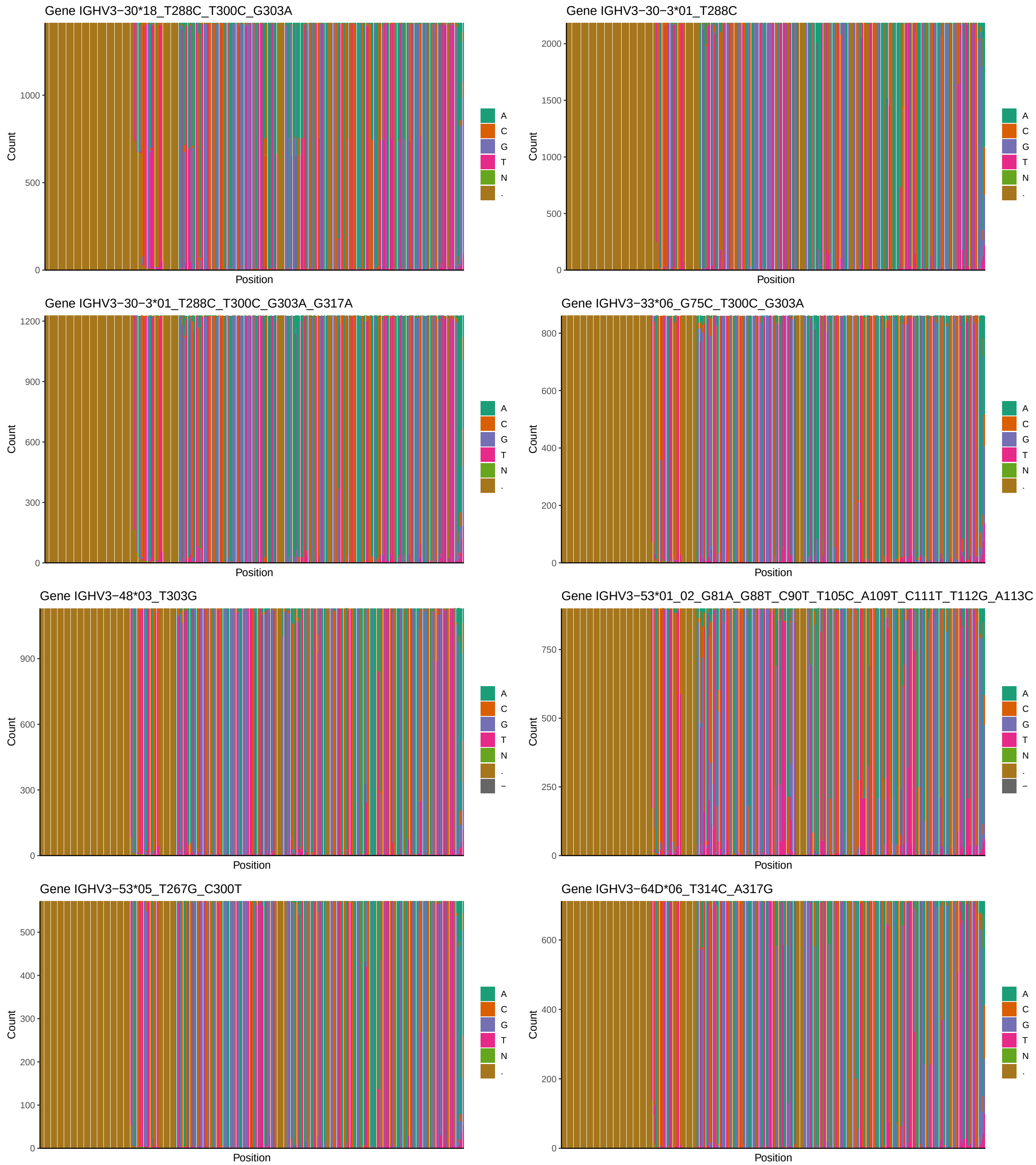








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1000
1500
2000
A
C
G
T
N
.
Gene IGHV3−30−3*01_T288C_T300C_G303A_G317A
Count
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600
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1200
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C
G
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Gene IGHV3−33*06_G75C_T300C_G303A
Count
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N
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Gene IGHV3−48*03_T303G
Count
Position
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A
C
G
T
N
.
−
Gene IGHV3−53*01_02_G81A_G88T_C90T_T105C_A109T_C111T_T112G_A113C
Count
Position
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C
G
T
N
.
−
Gene IGHV3−53*05_T267G_C300T
Count
Position
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C
G
T
N
.
Gene IGHV3−64D*06_T314C_A317G
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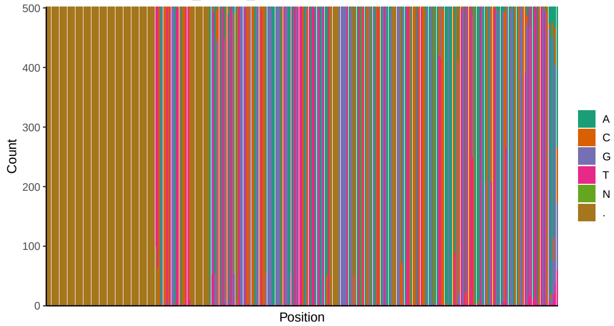
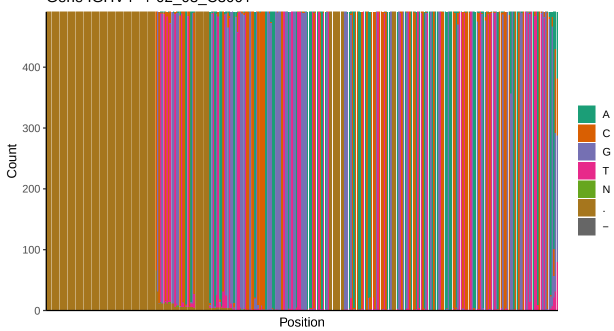
.

Gene IGHV3−64D*09_T314C_A317G



Gene IGHV3−74*01_A315G

Gene IGHV4−4*02_03_C300T



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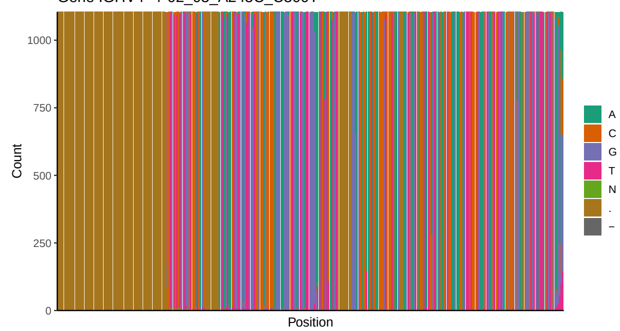

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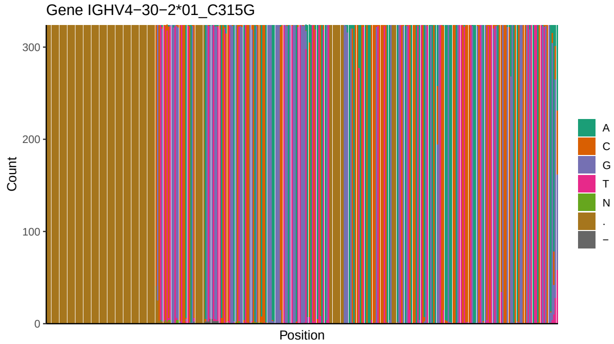

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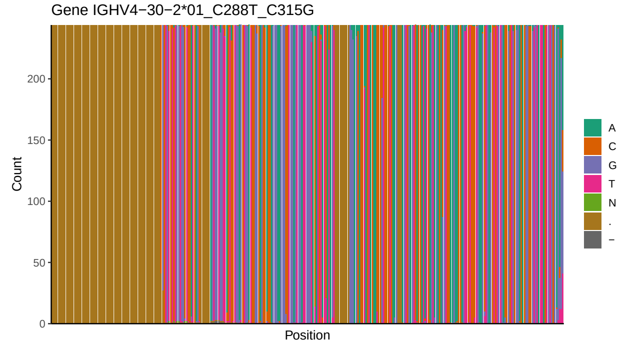

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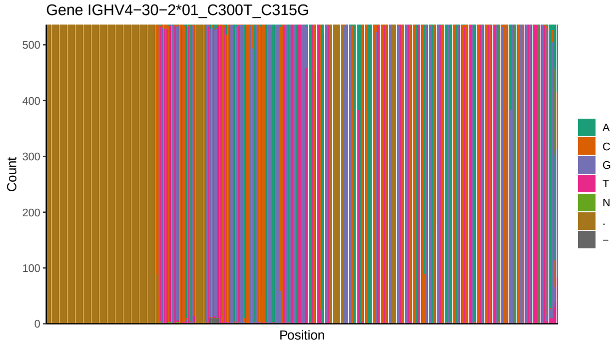

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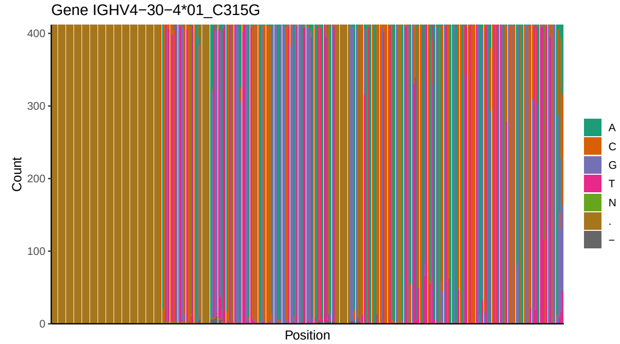

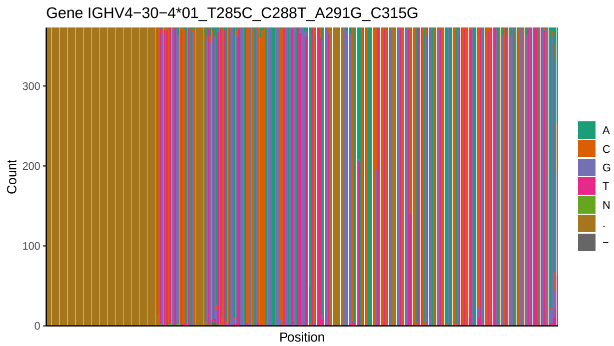
Gene IGHV4−30−4*01_T285C_C288T_A291G_C315G
Count
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100
200
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Position
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C
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T
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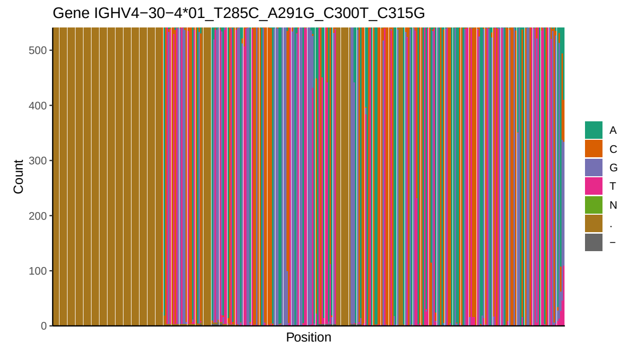
.
−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300
400
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Position
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G
T
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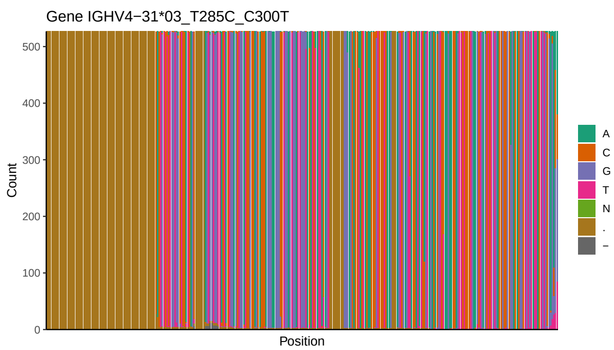
.
−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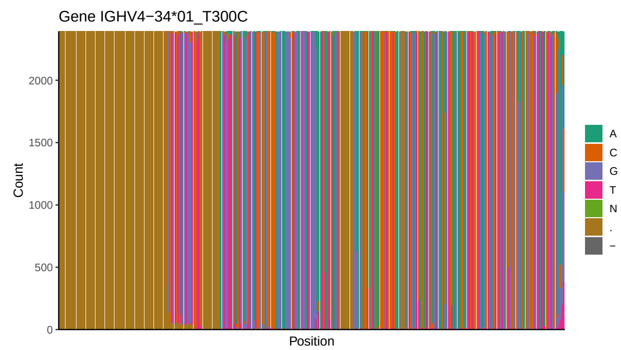
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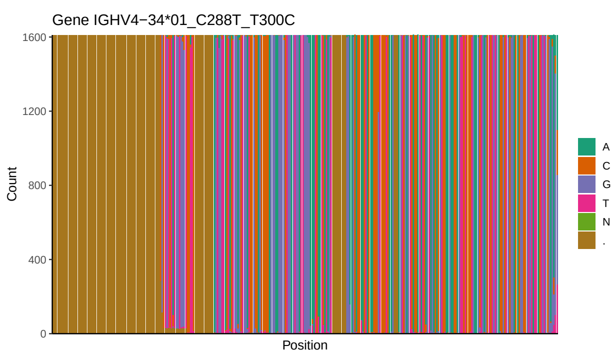
C
G
T
N
.
−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Position
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G
T
N
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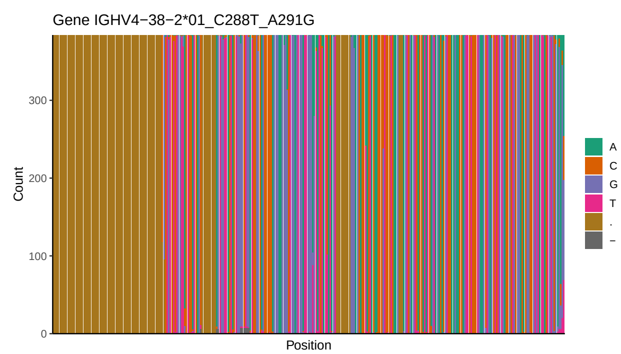
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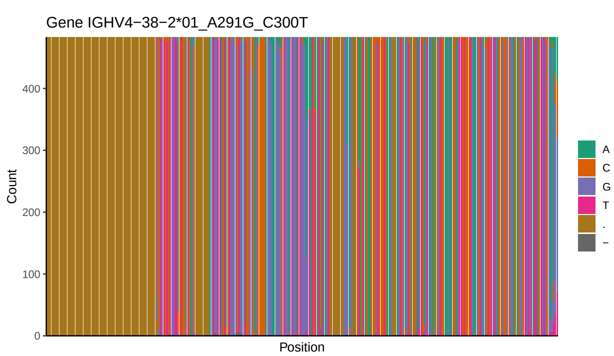
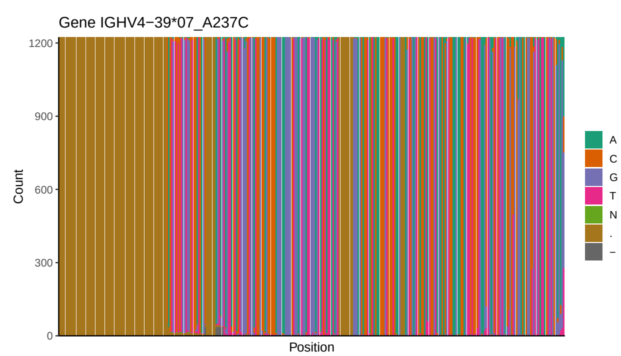
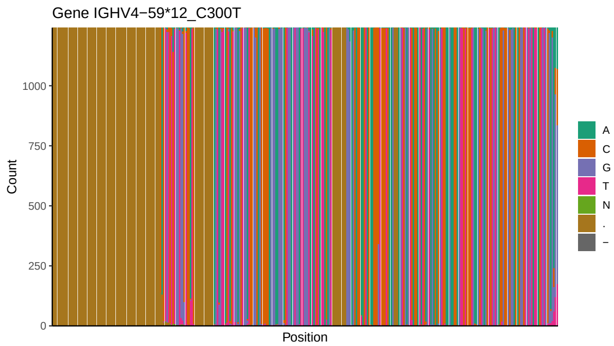
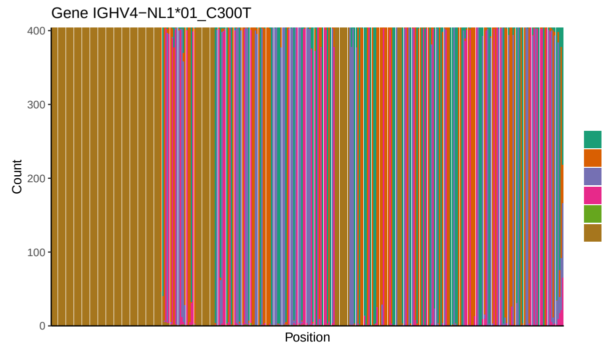
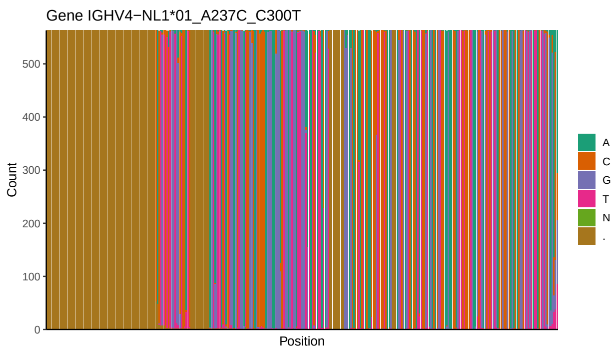
Position
A
C
G
T
.
−


Gene IGHV4−38−2*01_A291G_C300T
Count
0
100
200
300
400
Position
A
C
G
T
.
−

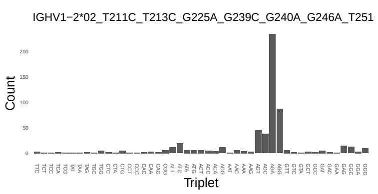

Gene IGHV4−39*07_A237C
Count
0
300
600
900
1200
Position
A
C
G
T
N
.
−

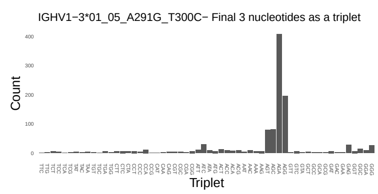


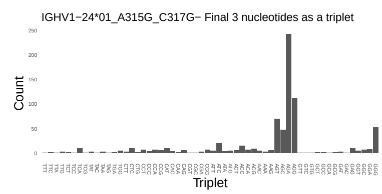


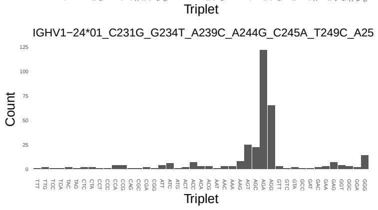


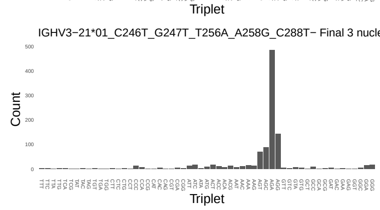
1.4 Final three nucleotides: frequency of each observed triplet

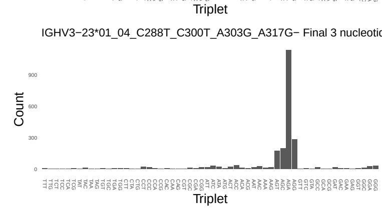


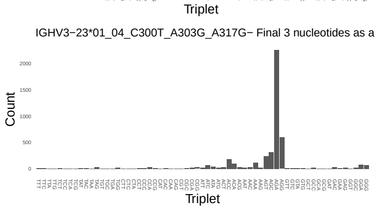


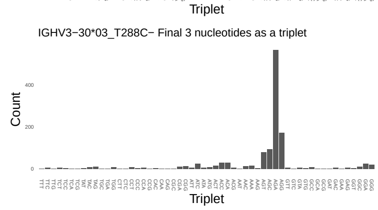


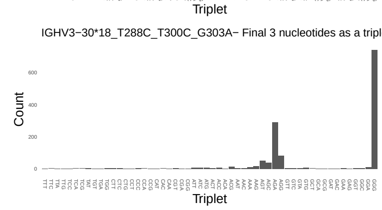


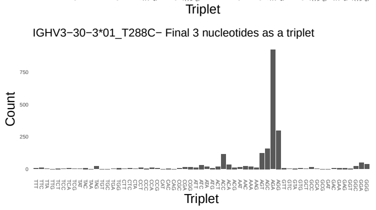


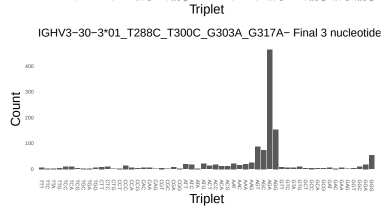


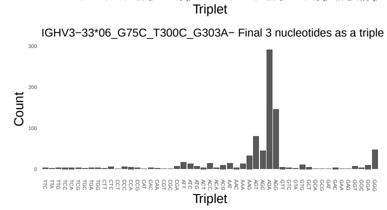


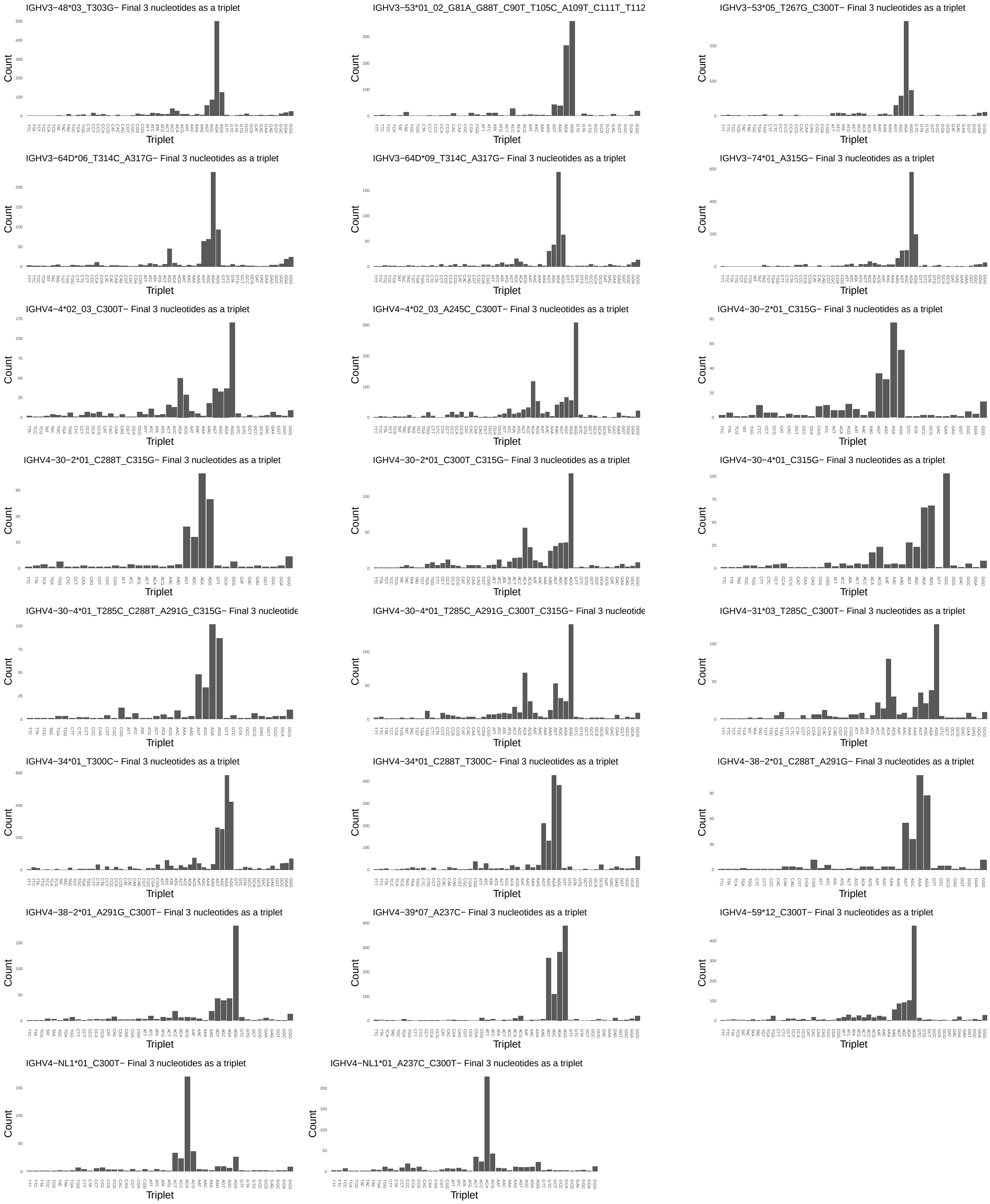




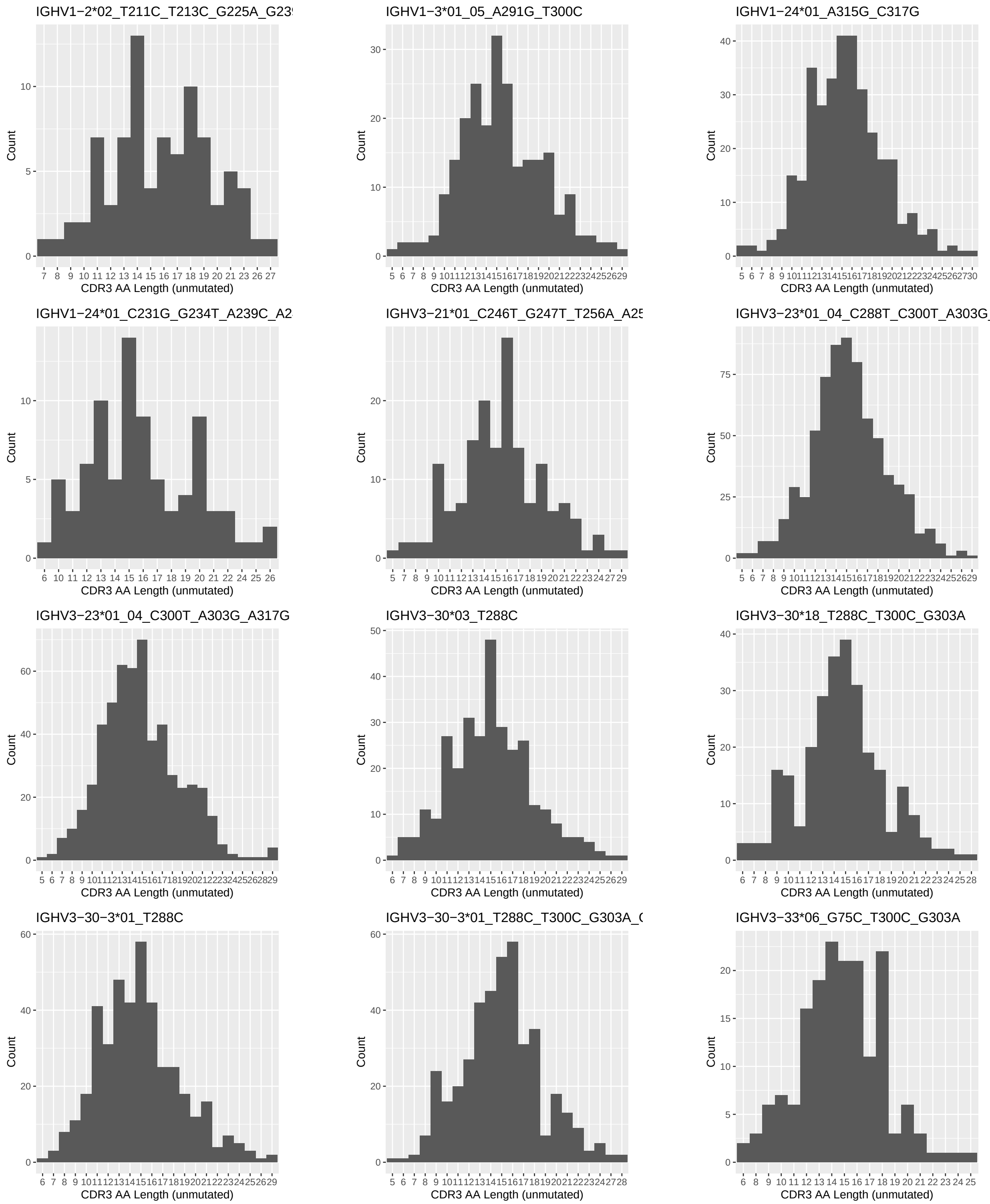


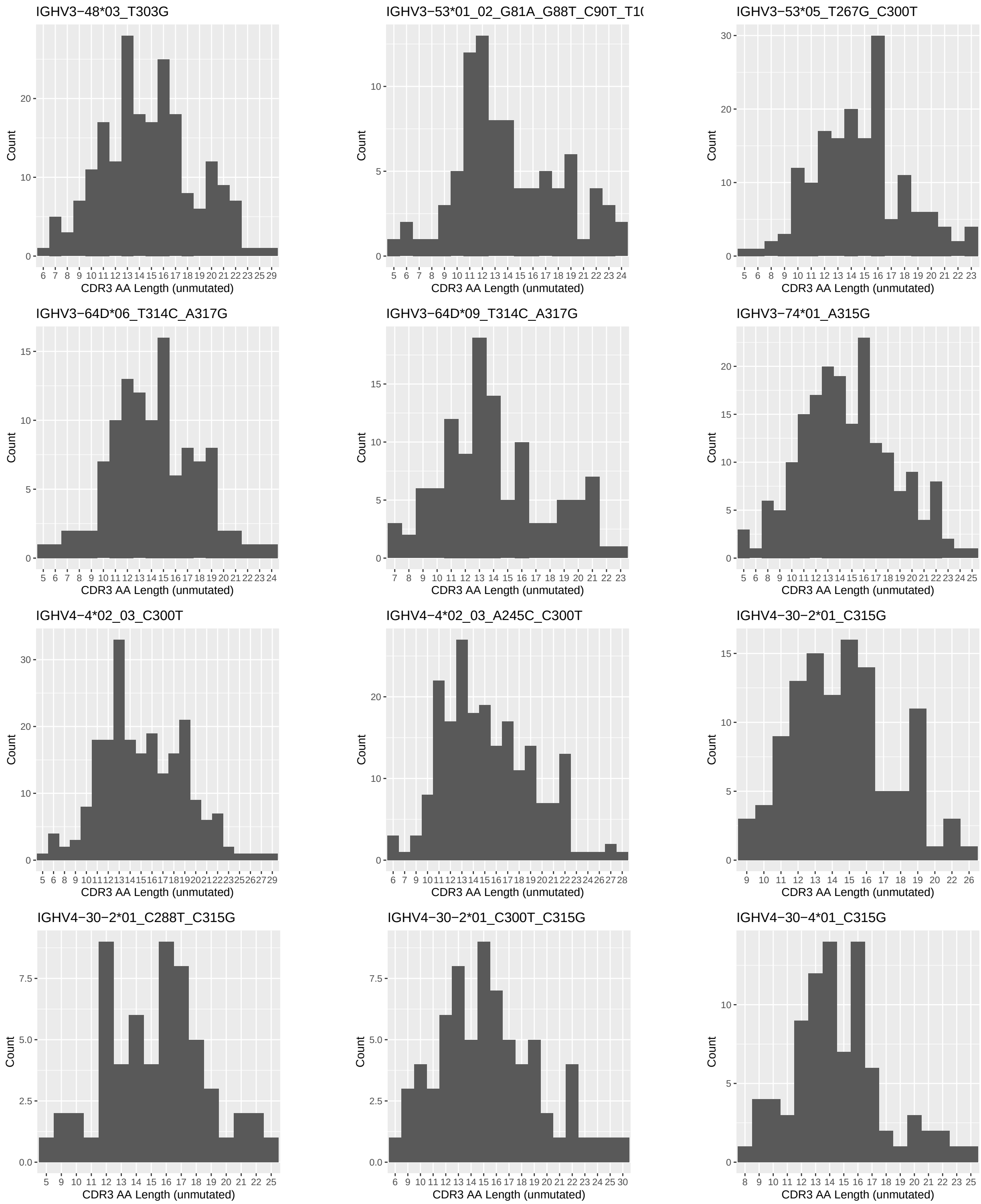


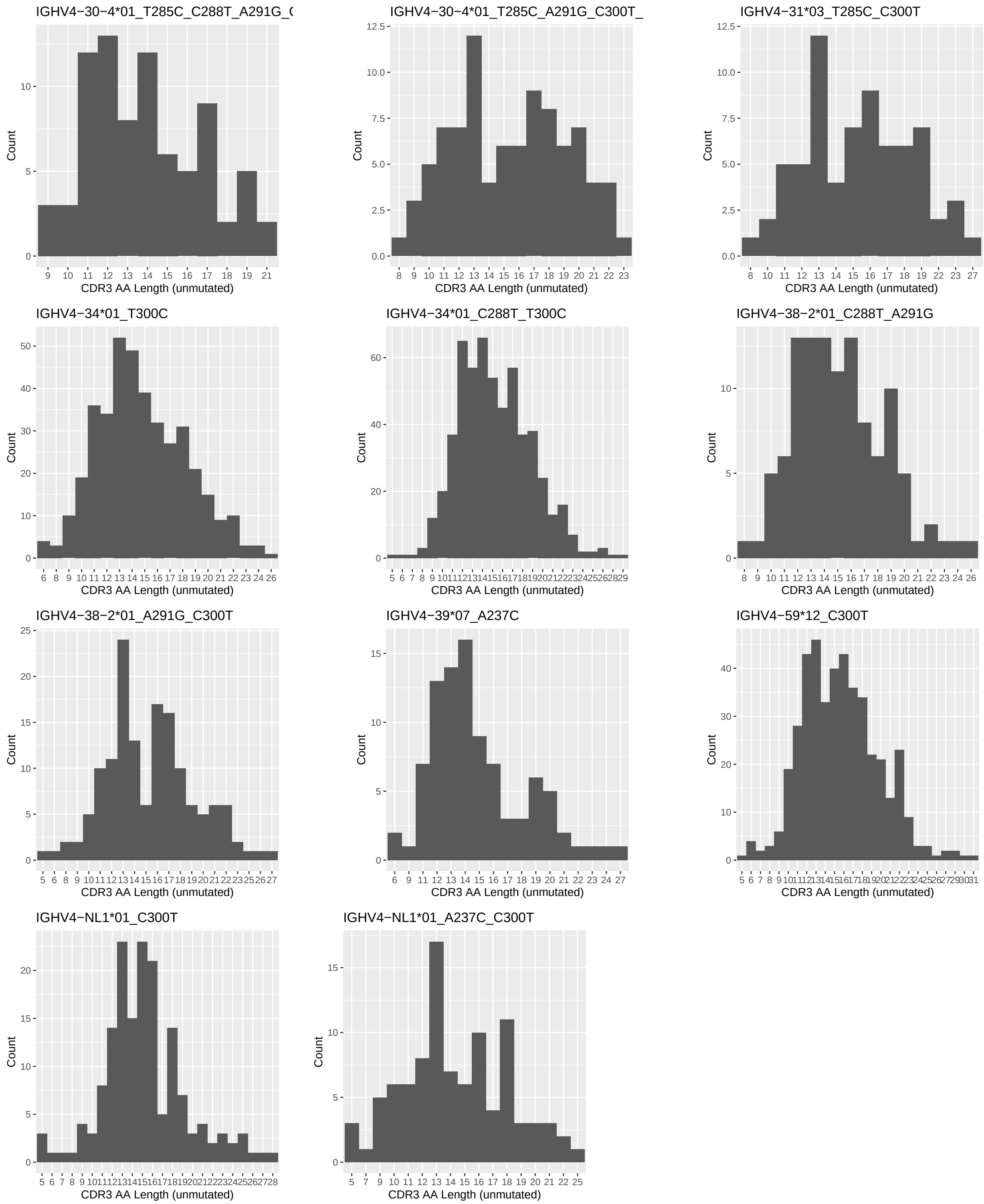



IGHV3−48*03_T303G− Final 3 nucleotides as a triplet
IGHV3−53*01_02_G81A_G88T_C90T_T105C_A109T_C111T_T112
IGHV3−53*05_T267G_C300T− Final 3 nucleotides as a triplet
IGHV3−64D*06_T314C_A317G− Final 3 nucleotides as a triplet
IGHV3−64D*09_T314C_A317G− Final 3 nucleotides as a triplet
IGHV3−74*01_A315G− Final 3 nucleotides as a triplet
IGHV4−4*02_03_C300T− Final 3 nucleotides as a triplet
IGHV4−4*02_03_A245C_C300T− Final 3 nucleotides as a triplet
IGHV4−30−2*01_C315G− Final 3 nucleotides as a triplet
IGHV4−30−2*01_C288T_C315G− Final 3 nucleotides as a triplet
IGHV4−30−2*01_C300T_C315G− Final 3 nucleotides as a triplet
IGHV4−30−4*01_C315G− Final 3 nucleotides as a triplet
IGHV4−30−4*01_T285C_C288T_A291G_C315G− Final 3 nucleotide
IGHV4−30−4*01_T285C_A291G_C300T_C315G− Final 3 nucleotide
IGHV4−31*03_T285C_C300T− Final 3 nucleotides as a triplet
IGHV4−34*01_T300C− Final 3 nucleotides as a triplet
IGHV4−34*01_C288T_T300C− Final 3 nucleotides as a triplet
IGHV4−38−2*01_C288T_A291G− Final 3 nucleotides as a triplet
IGHV4−38−2*01_A291G_C300T− Final 3 nucleotides as a triplet
IGHV4−39*07_A237C− Final 3 nucleotides as a triplet
IGHV4−59*12_C300T− Final 3 nucleotides as a triplet
IGHV4−NL1*01_C300T− Final 3 nucleotides as a triplet
IGHV4−NL1*01_A237C_C300T− Final 3 nucleotides as a triplet
Count
Triplet

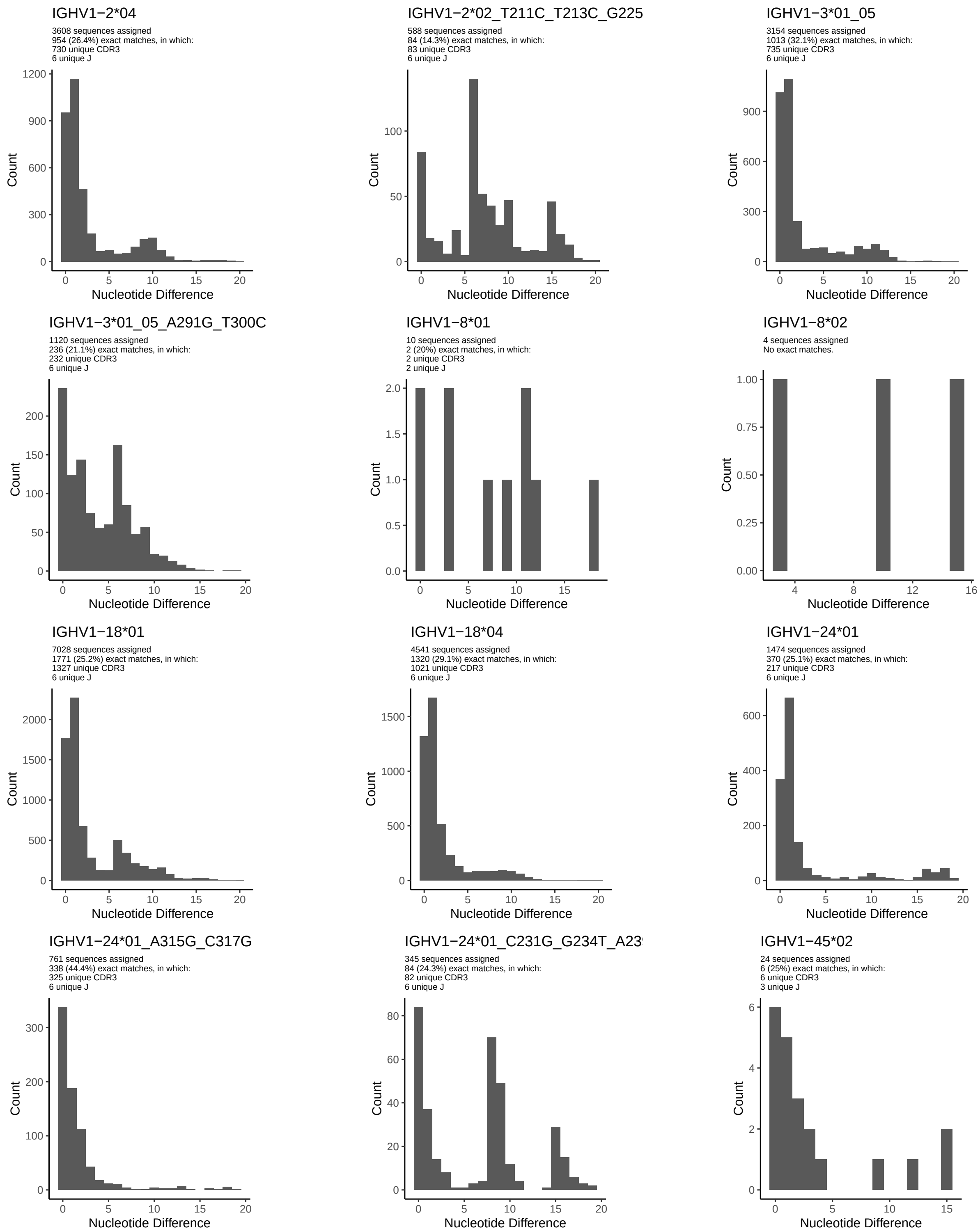
1.5 CDR3 length distribution, in assignments to novel alleles

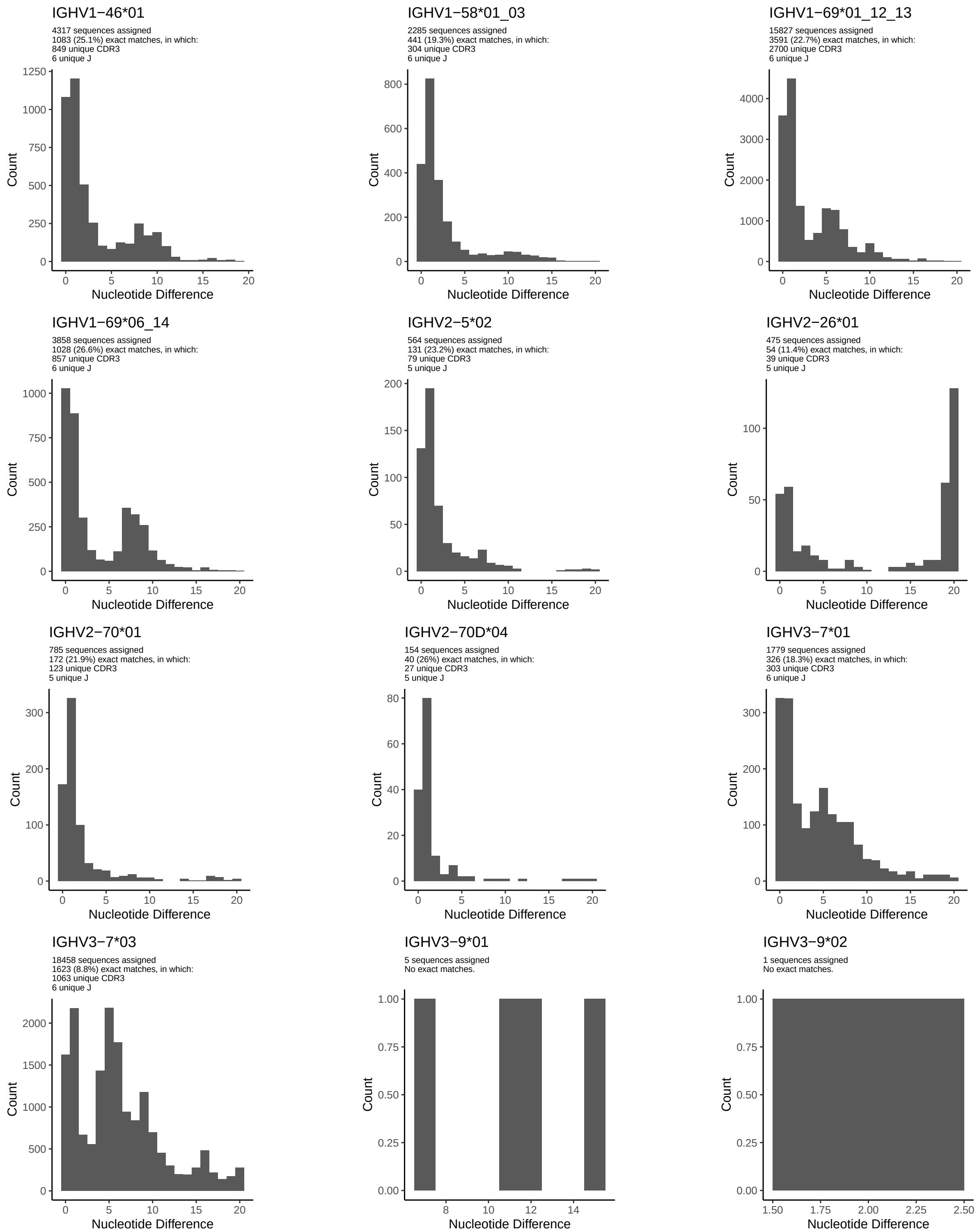


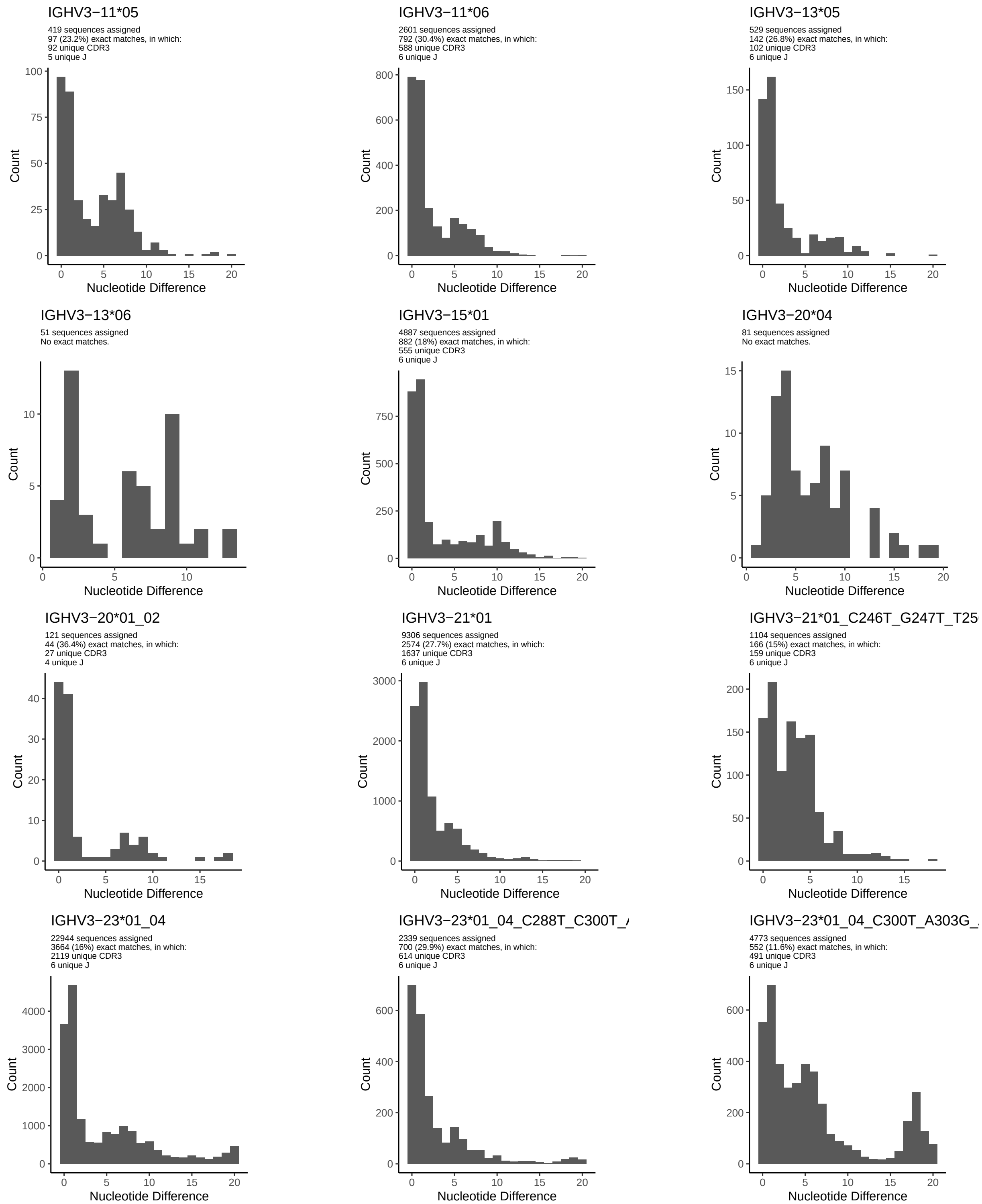

IGHV3−48*03_T303G
IGHV3−53*01_02_G81A_G88T_C90T_T1(
IGHV3−53*05_T267G_C300T
IGHV3−64D*06_T314C_A317G
IGHV3−64D*09_T314C_A317G
IGHV3−74*01_A315G
IGHV4−4*02_03_C300T
IGHV4−4*02_03_A245C_C300T
IGHV4−30−2*01_C315G
IGHV4−30−2*01_C288T_C315G
IGHV4−30−2*01_C300T_C315G
IGHV4−30−4*01_C315G
Count
CDR3 AA Length (unmutated)

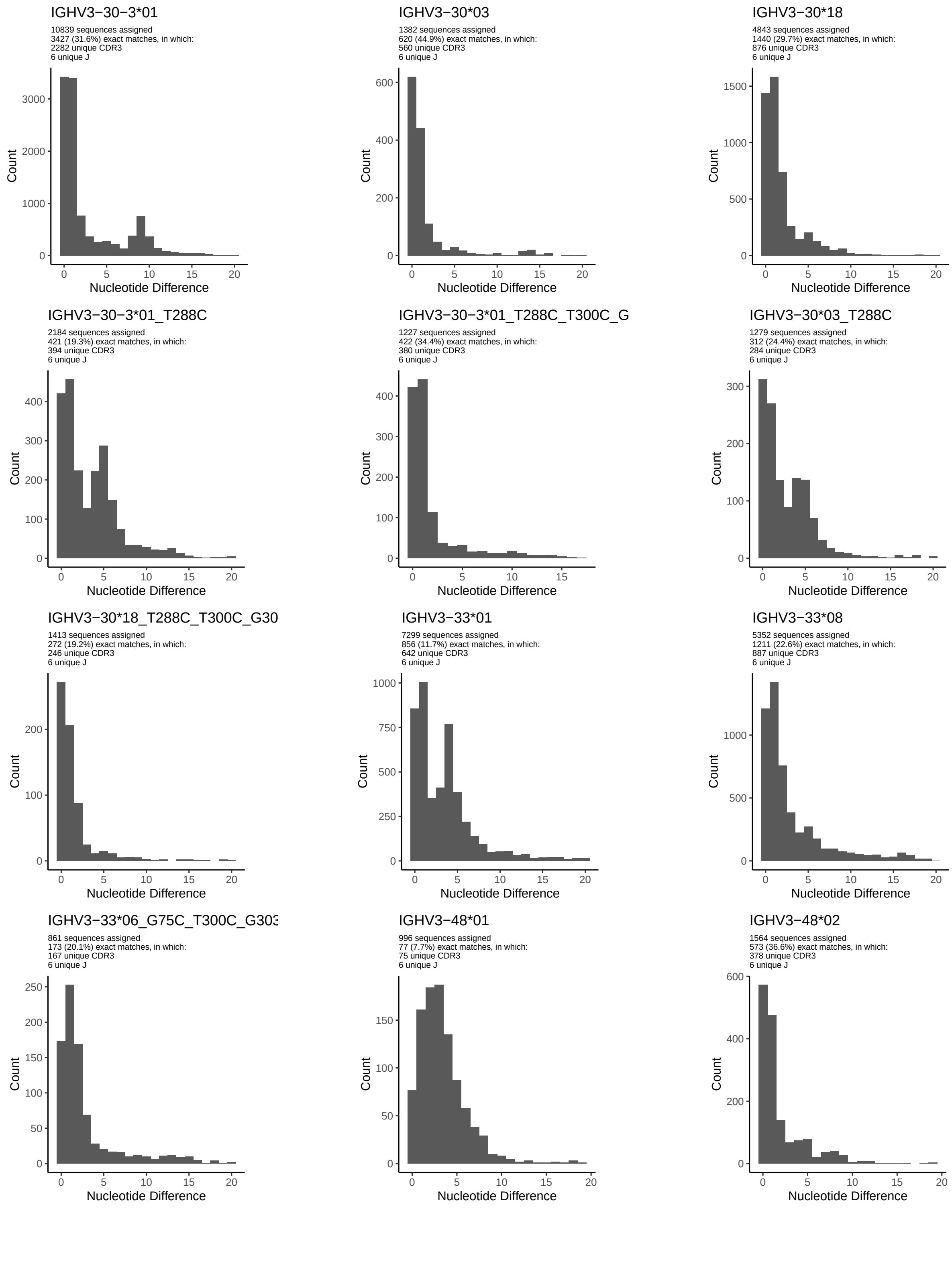

IGHV4−30−4*01_T285C_C288T_A291G_(
IGHV4−30−4*01_T285C_A291G_C300T_
IGHV4−31*03_T285C_C300T
IGHV4−34*01_T300C
IGHV4−34*01_C288T_T300C
IGHV4−38−2*01_C288T_A291G
IGHV4−38−2*01_A291G_C300T
IGHV4−39*07_A237C
IGHV4−59*12_C300T
IGHV4−NL1*01_C300T
IGHV4−NL1*01_A237C_C300T
Count
CDR3 AA Length (unmutated)

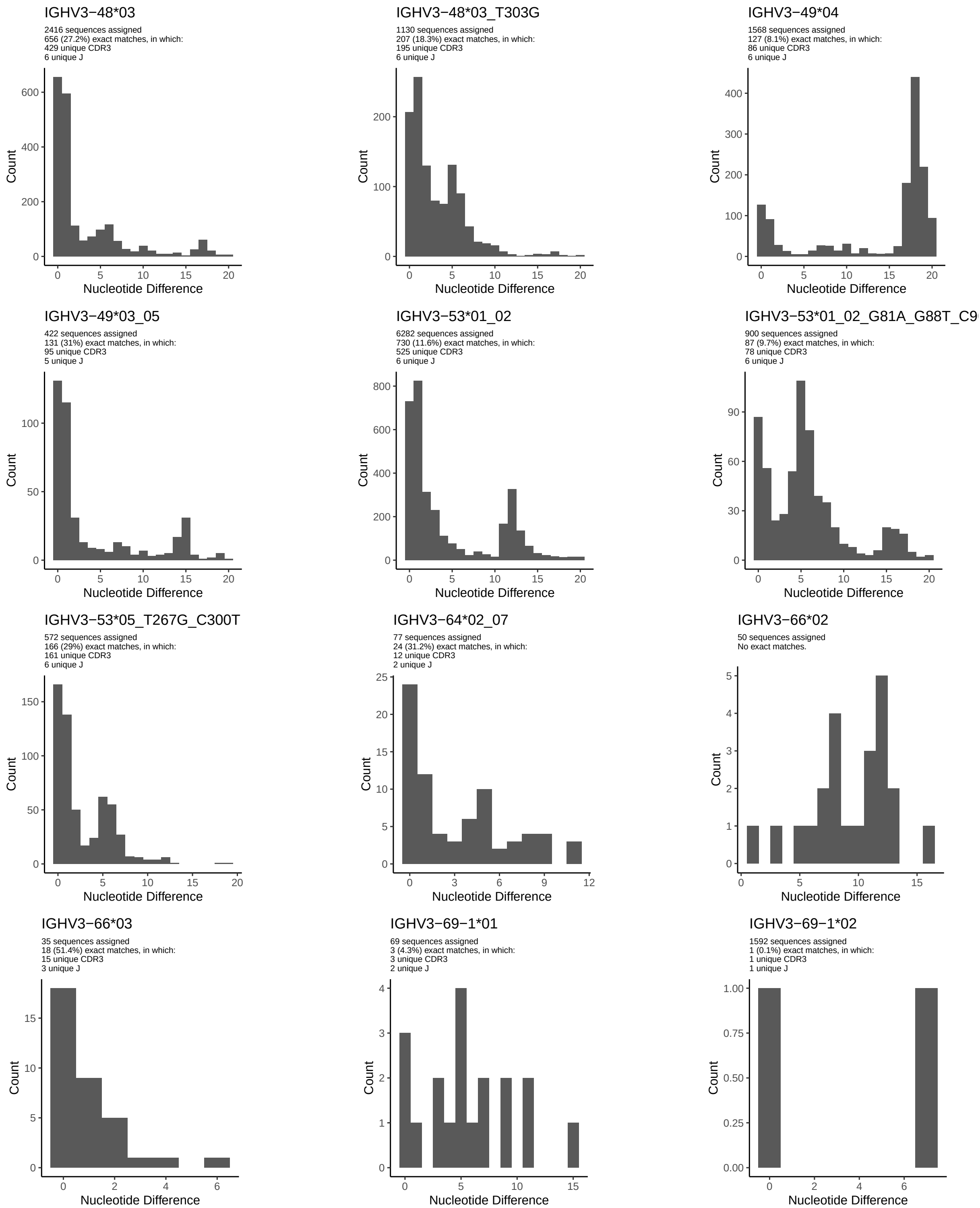
2 Variation from germline, in assignments to each allele

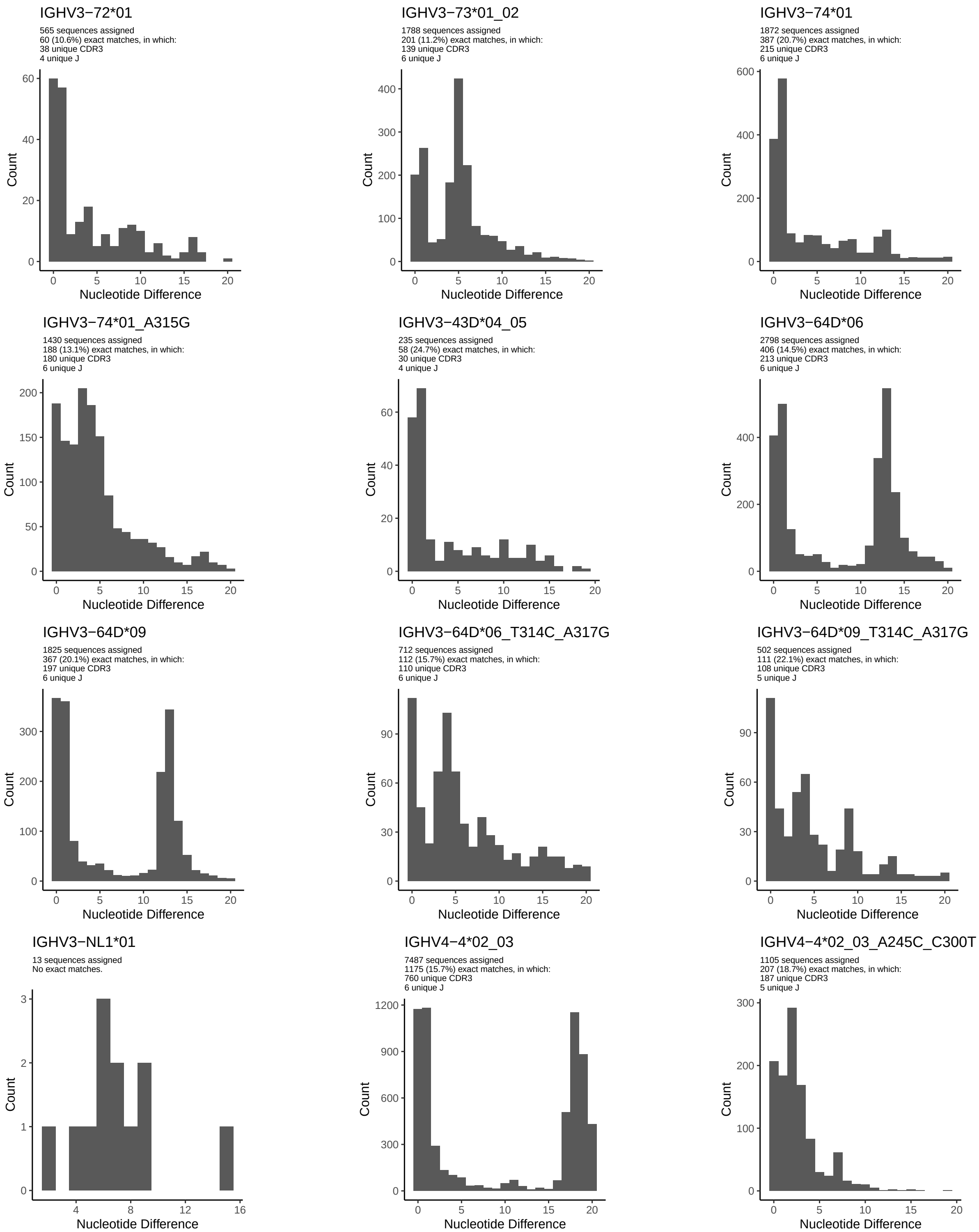


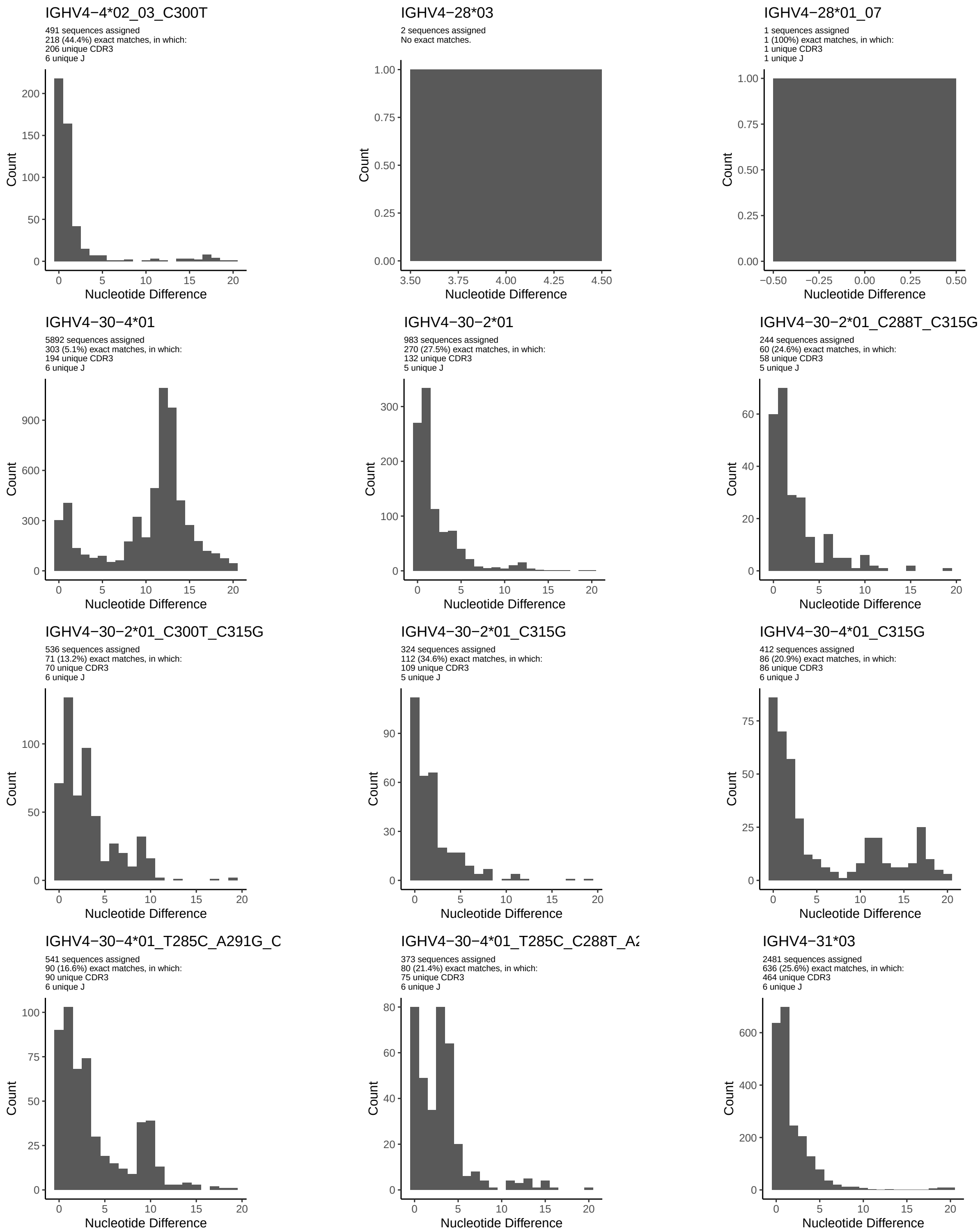

IGHV1−46*01
4317 sequences assigned
1083 (25.1%) exact matches, in which:
849 unique CDR3
6 unique J
IGHV1−58*01_03
2285 sequences assigned
441 (19.3%) exact matches, in which:
304 unique CDR3
6 unique J
IGHV1−69*01_12_13
15827 sequences assigned
3591 (22.7%) exact matches, in which:
2700 unique CDR3
6 unique J
IGHV1−69*06_14
3858 sequences assigned
1028 (26.6%) exact matches, in which:
857 unique CDR3
6 unique J
IGHV2−5*02
564 sequences assigned
131 (23.2%) exact matches, in which:
79 unique CDR3
5 unique J
IGHV2−26*01
475 sequences assigned
54 (11.4%) exact matches, in which:
39 unique CDR3
5 unique J
IGHV2−70*01
785 sequences assigned
172 (21.9%) exact matches, in which:
123 unique CDR3
5 unique J
IGHV2−70D*04
154 sequences assigned
40 (26%) exact matches, in which:
27 unique CDR3
5 unique J
IGHV3−7*01
1779 sequences assigned
326 (18.3%) exact matches, in which:
303 unique CDR3
6 unique J
IGHV3−7*03
18458 sequences assigned
1623 (8.8%) exact matches, in which:
1063 unique CDR3
6 unique J
IGHV3−9*01
5 sequences assigned
No exact matches.
IGHV3−9*02
1 sequences assigned
No exact matches.
Count
Nucleotide Difference

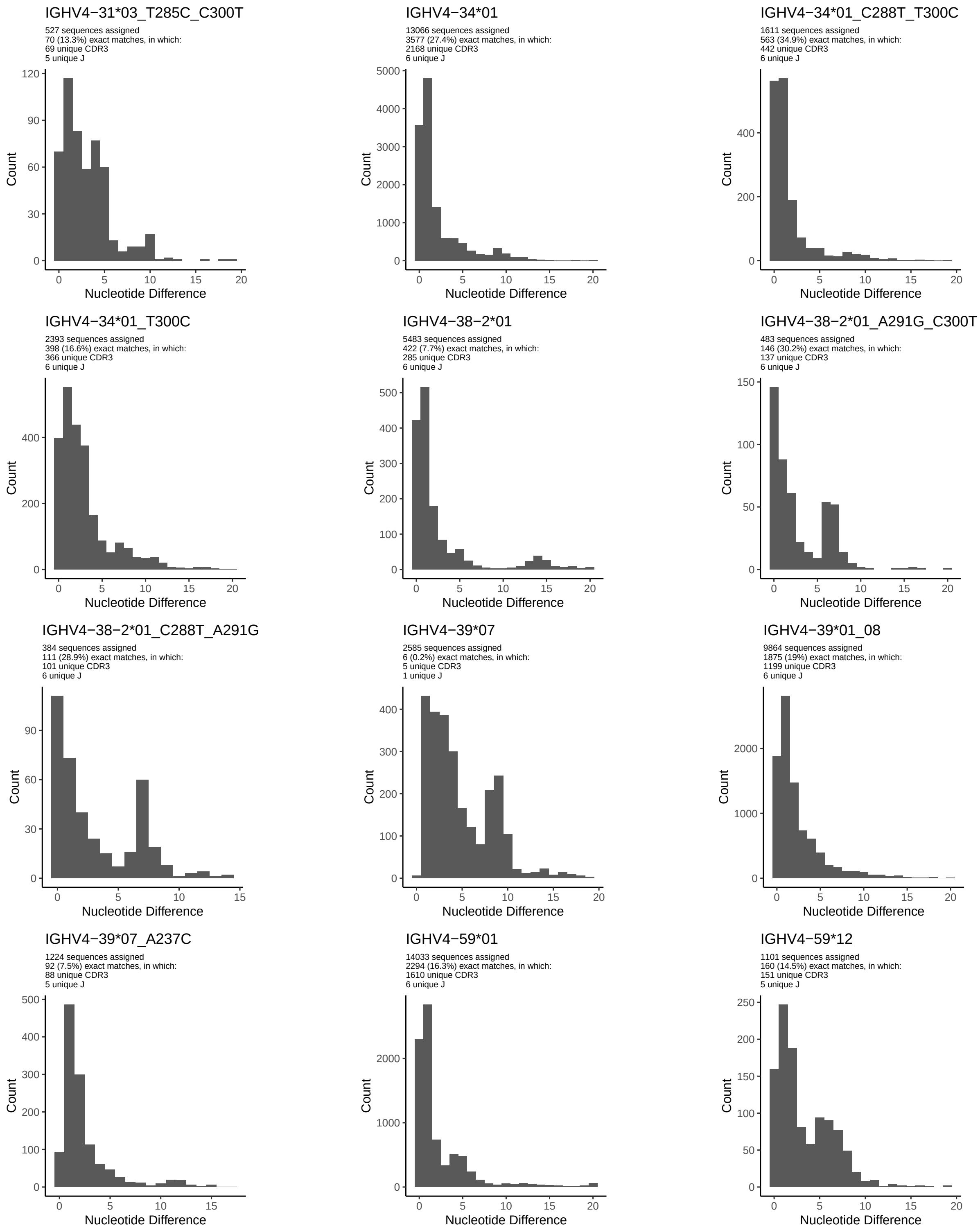

IGHV3−11*05
419 sequences assigned
97 (23.2%) exact matches, in which:
92 unique CDR3
5 unique J
IGHV3−11*06
2601 sequences assigned
792 (30.4%) exact matches, in which:
588 unique CDR3
6 unique J
IGHV3−13*05
529 sequences assigned
142 (26.8%) exact matches, in which:
102 unique CDR3
6 unique J
IGHV3−13*06
51 sequences assigned
No exact matches.
IGHV3−15*01
4887 sequences assigned
882 (18%) exact matches, in which:
555 unique CDR3
6 unique J
IGHV3−20*04
81 sequences assigned
No exact matches.
IGHV3−20*01_02
121 sequences assigned
44 (36.4%) exact matches, in which:
27 unique CDR3
4 unique J
IGHV3−21*01
9306 sequences assigned
2574 (27.7%) exact matches, in which:
1637 unique CDR3
6 unique J
IGHV3−21*01_C246T_G247T_T25
1104 sequences assigned
166 (15%) exact matches, in which:
159 unique CDR3
6 unique J
IGHV3−23*01_04
22944 sequences assigned
3664 (16%) exact matches, in which:
2119 unique CDR3
6 unique J
IGHV3−23*01_04_C288T_C300T_
2339 sequences assigned
700 (29.9%) exact matches, in which:
614 unique CDR3
6 unique J
IGHV3−23*01_04_C300T_A303G_
4773 sequences assigned
552 (11.6%) exact matches, in which:
491 unique CDR3
6 unique J
Count
Nucleotide Difference

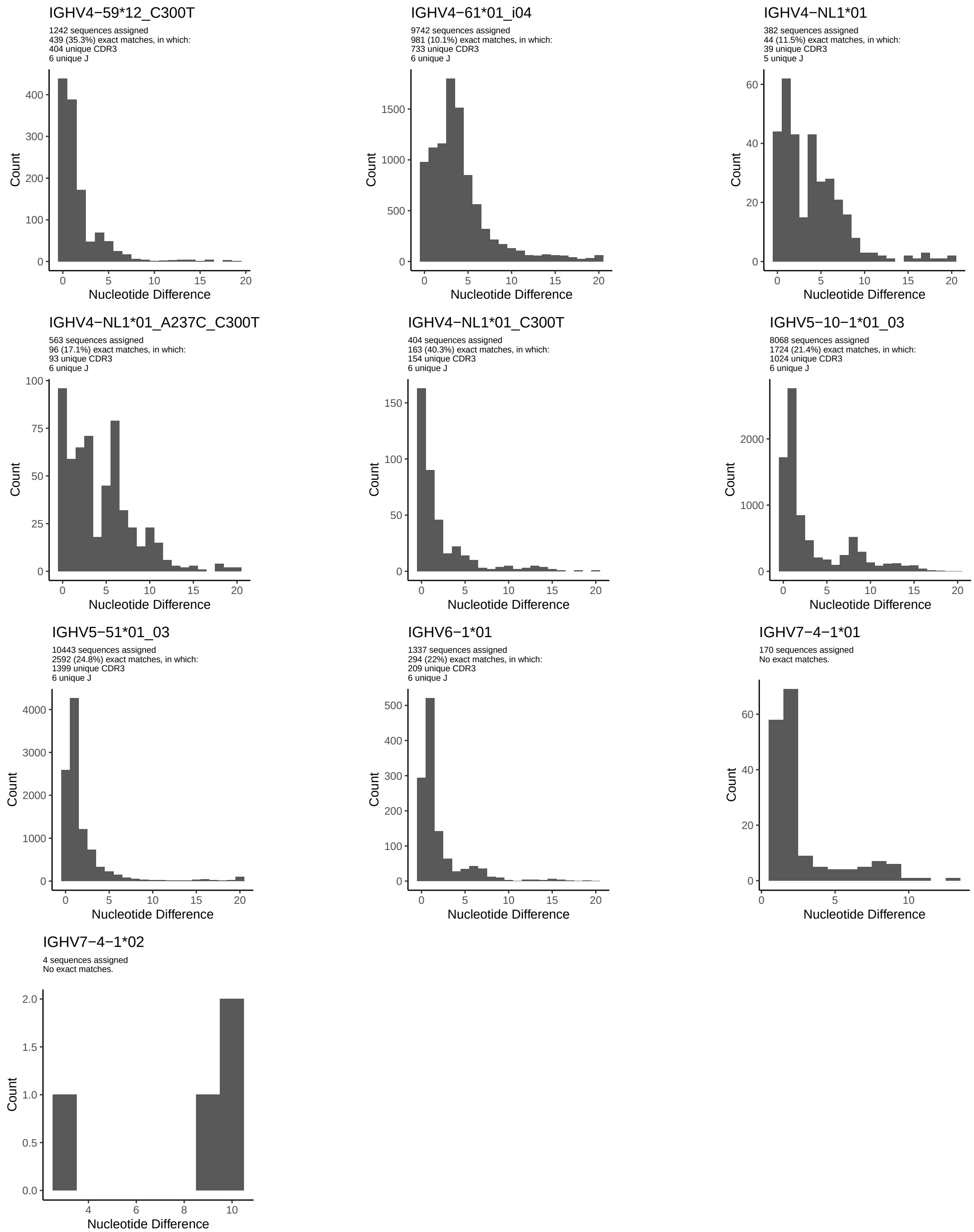

IGHV3−30−3*01
10839 sequences assigned
3427 (31.6%) exact matches, in which:
2282 unique CDR3
6 unique J
IGHV3−30*03
1382 sequences assigned
620 (44.9%) exact matches, in which:
560 unique CDR3
6 unique J
IGHV3−30*18
4843 sequences assigned
1440 (29.7%) exact matches, in which:
876 unique CDR3
6 unique J
IGHV3−30−3*01_T288C
2184 sequences assigned
421 (19.3%) exact matches, in which:
394 unique CDR3
6 unique J
IGHV3−30−3*01_T288C_T300C_G
1227 sequences assigned
422 (34.4%) exact matches, in which:
380 unique CDR3
6 unique J
IGHV3−30*03_T288C
1279 sequences assigned
312 (24.4%) exact matches, in which:
284 unique CDR3
6 unique J
IGHV3−30*18_T288C_T300C_G30
1413 sequences assigned
272 (19.2%) exact matches, in which:
246 unique CDR3
6 unique J
IGHV3−33*01
7299 sequences assigned
856 (11.7%) exact matches, in which:
642 unique CDR3
6 unique J
IGHV3−33*08
5352 sequences assigned
1211 (22.6%) exact matches, in which:
887 unique CDR3
6 unique J
IGHV3−33*06_G75C_T300C_G30
861 sequences assigned
173 (20.1%) exact matches, in which:
167 unique CDR3
6 unique J
IGHV3−48*01
996 sequences assigned
77 (7.7%) exact matches, in which:
75 unique CDR3
6 unique J
IGHV3−48*02
1564 sequences assigned
573 (36.6%) exact matches, in which:
378 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV3−48*03
2416 sequences assigned
656 (27.2%) exact matches, in which:
429 unique CDR3
6 unique J
IGHV3−48*03_T303G
1130 sequences assigned
207 (18.3%) exact matches, in which:
195 unique CDR3
6 unique J
IGHV3−49*04
1568 sequences assigned
127 (8.1%) exact matches, in which:
86 unique CDR3
6 unique J
IGHV3−49*03_05
422 sequences assigned
131 (31%) exact matches, in which:
95 unique CDR3
5 unique J
IGHV3−53*01_02
6282 sequences assigned
730 (11.6%) exact matches, in which:
525 unique CDR3
6 unique J
IGHV3−53*01_02_G81A_G88T_C9
900 sequences assigned
87 (9.7%) exact matches, in which:
78 unique CDR3
6 unique J
IGHV3−53*05_T267G_C300T
572 sequences assigned
166 (29%) exact matches, in which:
161 unique CDR3
6 unique J
IGHV3−64*02_07
77 sequences assigned
24 (31.2%) exact matches, in which:
12 unique CDR3
2 unique J
IGHV3−66*02
50 sequences assigned
No exact matches.
IGHV3−66*03
35 sequences assigned
18 (51.4%) exact matches, in which:
15 unique CDR3
3 unique J
IGHV3−69−1*01
69 sequences assigned
3 (4.3%) exact matches, in which:
3 unique CDR3
2 unique J
IGHV3−69−1*02
1592 sequences assigned
1 (0.1%) exact matches, in which:
1 unique CDR3
1 unique J
Count
Nucleotide Difference


IGHV3−72*01
565 sequences assigned
60 (10.6%) exact matches, in which:
38 unique CDR3
4 unique J
IGHV3−73*01_02
1788 sequences assigned
201 (11.2%) exact matches, in which:
139 unique CDR3
6 unique J
IGHV3−74*01
1872 sequences assigned
387 (20.7%) exact matches, in which:
215 unique CDR3
6 unique J
IGHV3−74*01_A315G
1430 sequences assigned
188 (13.1%) exact matches, in which:
180 unique CDR3
6 unique J
IGHV3−43D*04_05
235 sequences assigned
58 (24.7%) exact matches, in which:
30 unique CDR3
4 unique J
IGHV3−64D*06
2798 sequences assigned
406 (14.5%) exact matches, in which:
213 unique CDR3
6 unique J
IGHV3−64D*09
1825 sequences assigned
367 (20.1%) exact matches, in which:
197 unique CDR3
6 unique J
IGHV3−64D*06_T314C_A317G
712 sequences assigned
112 (15.7%) exact matches, in which:
110 unique CDR3
6 unique J
IGHV3−64D*09_T314C_A317G
502 sequences assigned
111 (22.1%) exact matches, in which:
108 unique CDR3
5 unique J
IGHV3−NL1*01
13 sequences assigned
No exact matches.
IGHV4−4*02_03
7487 sequences assigned
1175 (15.7%) exact matches, in which:
760 unique CDR3
6 unique J
IGHV4−4*02_03_A245C_C300T
1105 sequences assigned
207 (18.7%) exact matches, in which:
187 unique CDR3
5 unique J
Count
Nucleotide Difference

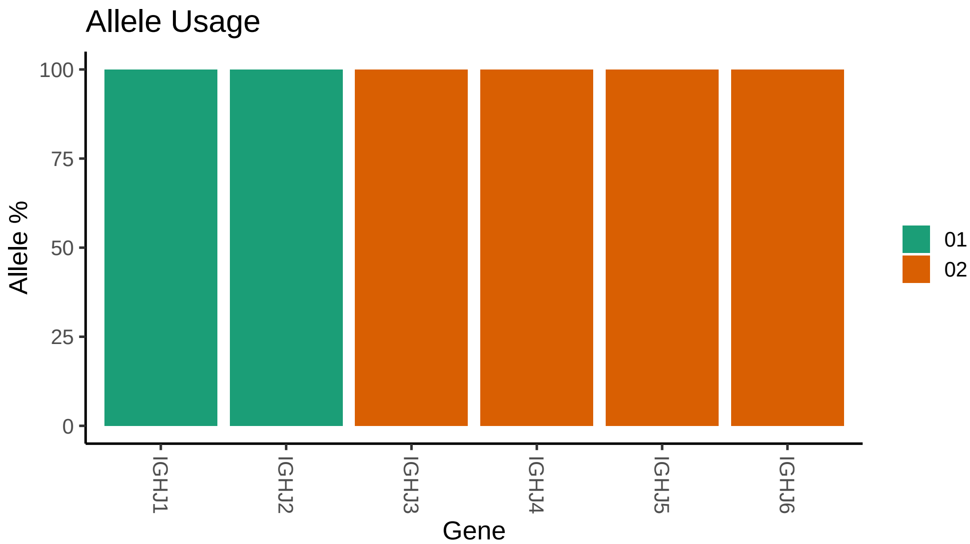

IGHV4−4*02_03_C300T
491 sequences assigned
218 (44.4%) exact matches, in which:
206 unique CDR3
6 unique J
IGHV4−28*03
2 sequences assigned
No exact matches.
IGHV4−28*01_07
1 sequences assigned
1 (100%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV4−30−4*01
5892 sequences assigned
303 (5.1%) exact matches, in which:
194 unique CDR3
6 unique J
IGHV4−30−2*01
983 sequences assigned
270 (27.5%) exact matches, in which:
132 unique CDR3
5 unique J
IGHV4−30−2*01_C288T_C315G
244 sequences assigned
60 (24.6%) exact matches, in which:
58 unique CDR3
5 unique J
IGHV4−30−2*01_C300T_C315G
536 sequences assigned
71 (13.2%) exact matches, in which:
70 unique CDR3
6 unique J
IGHV4−30−2*01_C315G
324 sequences assigned
112 (34.6%) exact matches, in which:
109 unique CDR3
5 unique J
IGHV4−30−4*01_C315G
412 sequences assigned
86 (20.9%) exact matches, in which:
86 unique CDR3
6 unique J
IGHV4−30−4*01_T285C_A291G_C
541 sequences assigned
90 (16.6%) exact matches, in which:
90 unique CDR3
6 unique J
IGHV4−30−4*01_T285C_C288T_A
373 sequences assigned
80 (21.4%) exact matches, in which:
75 unique CDR3
6 unique J
IGHV4−31*03
2481 sequences assigned
636 (25.6%) exact matches, in which:
464 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−31*03_T285C_C300T
527 sequences assigned
70 (13.3%) exact matches, in which:
69 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV4−34*01
13066 sequences assigned
3577 (27.4%) exact matches, in which:
2168 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−34*01_C288T_T300C
1611 sequences assigned
563 (34.9%) exact matches, in which:
442 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−34*01_T300C
2393 sequences assigned
398 (16.6%) exact matches, in which:
366 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−38−2*01
5483 sequences assigned
422 (7.7%) exact matches, in which:
285 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−38−2*01_A291G_C300T
483 sequences assigned
146 (30.2%) exact matches, in which:
137 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−38−2*01_C288T_A291G
384 sequences assigned
111 (28.9%) exact matches, in which:
101 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−39*07
2585 sequences assigned
6 (0.2%) exact matches, in which:
5 unique CDR3
1 unique J
Count
Nucleotide Difference
IGHV4−39*01_08
9864 sequences assigned
1875 (19%) exact matches, in which:
1199 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−39*07_A237C
1224 sequences assigned
92 (7.5%) exact matches, in which:
88 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV4−59*01
14033 sequences assigned
2294 (16.3%) exact matches, in which:
1610 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−59*12
1101 sequences assigned
160 (14.5%) exact matches, in which:
151 unique CDR3
5 unique J
Count
Nucleotide Difference



IGHV4−59*12_C300T
1242 sequences assigned
439 (35.3%) exact matches, in which:
404 unique CDR3
6 unique J
IGHV4−61*01_i04
9742 sequences assigned
981 (10.1%) exact matches, in which:
733 unique CDR3
6 unique J
IGHV4−NL1*01
382 sequences assigned
44 (11.5%) exact matches, in which:
39 unique CDR3
5 unique J
IGHV4−NL1*01_A237C_C300T
563 sequences assigned
96 (17.1%) exact matches, in which:
93 unique CDR3
6 unique J
IGHV4−NL1*01_C300T
404 sequences assigned
163 (40.3%) exact matches, in which:
154 unique CDR3
6 unique J
IGHV5−10−1*01_03
8068 sequences assigned
1724 (21.4%) exact matches, in which:
1024 unique CDR3
6 unique J
IGHV5−51*01_03
10443 sequences assigned
2592 (24.8%) exact matches, in which:
1399 unique CDR3
6 unique J
IGHV6−1*01
1337 sequences assigned
294 (22%) exact matches, in which:
209 unique CDR3
6 unique J
IGHV7−4−1*01
170 sequences assigned
No exact matches.
IGHV7−4−1*02
4 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /work/jenkins/PRJNA491287/M64 050/M64 050/M64 050/M64 050/18/9106384be6e5ae724ba1c7
##
## Germline reference file: /work/jenkins/PRJNA491287/M64 050/M64 050/M64 050/M64 050/18/9106384be6e5ae
##
## Novel allele file: /work/jenkins/PRJNA491287/M64 050/M64 050/M64 050/M64 050/18/9106384be6e5ae724ba1
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 2 04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 2 02_T211C_T213C_G225A_G239C_G240A_G246A_T251C_C252G_C276G_G279
##
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_C231G_G234T_A239C_A244G_C245A_T249C_A252G_G253A_A254G_A31
##
##
## Unexpected N in reference sequence IGHV3 21 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 21 01_C246T_G247T_T256A_A258G_C288T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C288T_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
```

```
##
##
## Unexpected N in reference sequence IGHV3 30 18: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 18_T288C_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C_T300C_G303A_G317A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 08: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_G75C_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 48 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 03_T303G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 53 01_02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 53 01_02_G81A_G88T_C90T_T105C_A109T_C111T_T112G_A113C: replacin
##
##
## Unexpected N in reference sequence IGHV3 66 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 53 05_T267G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 64D 06: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 64D 06_T314C_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 64D 09: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 64D 09_T314C_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C288T_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C300T_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 4 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_T285C_C288T_A291G_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_T285C_A291G_C300T_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 31 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 31 03_T285C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_C288T_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 38 2 01: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV4 38 2 01_C288T_A291G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 01_A291G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 39 07: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 39 07_A237C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_A237C_C300T: replacing with gap
```